

OISD – STD - 129

Revised Edition

FOR RESTRICTED CIRCULATION

INSPECTION OF STORAGE TANKS

OISD - STANDARD-129

First Edition, November 1988

Amended Edition, August 1999

Revised Edition, November 2006



Oil Industry Safety Directorate

Government of India

Ministry of Petroleum & Natural Gas

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No.

INSPECTION OF STORAGE TANKS

Prepared by

Functional Committee

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Preamble

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for sovereignty and prosperity of the country. As the industry essentially deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as only safe performance at all times can ensure optimum ROI of these national assets and resources including sustainability.

While statutory organizations were in place all along to oversee safety aspects of Indian petroleum industry, Oil Industry Safety Directorate (OISD) was set up in 1986 Ministry of Petroleum and Natural Gas, Government of India as a knowledge centre for formulation of constantly updated world-scale standards for design, layout and operation of various equipment, facility and activities involved in this industry. Moreover, OISD was also given responsibility of monitoring implementation status of these standards through safety audits.

In more than 25 years of its existence, OISD has developed a rigorous, multi-layer, iterative and participative process of development of standards – starting with research by in-house experts and iterating through seeking & validating inputs from all stake-holders – operators, designers, national level knowledge authorities and public at large – with a feedback loop of constant updation based on ground level experience obtained through audits, incident analysis and environment scanning.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry culminating in a safer environment in the industry. OISD – except in the Upstream Petroleum Sector – is still a regulatory (and not a statutory) body but that has not affected implementation of the OISD standards. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption in various statutory rules of the land.

Petroleum industry in India is significantly globalized at present in terms of technology content requiring its operation to keep pace with the relevant world scale standards & practices. This matches the OISD philosophy of continuous improvement keeping pace with the global developments in its target environment. To this end, OISD keeps track of changes through participation as member in large number of International and national level Knowledge Organizations – both in the field of standard development and implementation & monitoring in addition to updation of internal knowledge base through continuous research and application surveillance, thereby ensuring that this OISD Standard, along with all other extant ones, remains relevant, updated and effective on a real time basis in the applicable areas.

Together we strive to achieve NIL incidents in the entire Hydrocarbon Value Chain. This, besides other issues, calls for total engagement from all levels of the stake holder organizations, which we, at OISD, fervently look forward to.

Jai Hind!!!

Executive Director

Oil Industry Safety Directorate

FOREWORD

The Oil Industry in India is 100 years old. Because of various collaboration agreements, a variety of international codes, standards and practices have been in vogue. Standardisation in design philosophies and operating and maintenance practices at a national level was hardly in existence. This, coupled with feedback from some serious accidents that occurred in the recent past in India and abroad, emphasized the need for the industry to review the existing state of art in designing, operating and maintaining oil and gas installations.

With this in view, the Ministry of Petroleum & Natural Gas, in 1986, constituted a Safety Council assisted by Oil Industry Safety Directorate (OISD) staffed from within the industry in formulating and implementing a series of self-regulatory measures aimed at removing obsolescence, standardising and upgrading the existing standards to ensure safe operations. Accordingly, OISD constituted a number of functional committees comprising of experts nominated from the industry to draw up standards and guidelines on various subjects.

The present document on "Inspection of Storage Tanks" was prepared by the Functional Committee on "Inspection of Static Equipment". This document is based on the accumulated knowledge and experience of industry members and the various national and international codes and practices. This document is meant to be used as a supplement and not as a replacement for existing codes and standards. It is hoped that the provisions of this document, when adopted may go a long way to improve the safety and reduce accidents in the oil and gas Industry. Users of this document are cautioned that no standard can be a substitute for a responsible qualified Inspection Engineers. Suggestions are invited from the users after it is put into practice to improve the document further.

This standard in no way supersedes the statutory regulations of CCE, Factory Inspectorate or other Government bodies, which must be followed as applicable.

Suggestions for amendments to this document should be addressed to

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These documents are intended only to supplement and not replace the prevailing statutory requirements.

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INSPECTION OF STORAGE TANKS

1.0 INTRODUCTION

Petroleum products, allied chemicals and water are stored in tanks as finished inventory/ intermediate products/ feed in petroleum refineries, product depots etc. Storage tanks of various types and sizes are used to store these products. The failure of any part of a tank may lead to a disaster.

Timely inspection and preventive maintenance of these storage tanks assume high importance. Accordingly, the inspection schedules of storage tanks are to be prepared and implemented.

2.0. SCOPE

This standard covers the minimum inspection requirements for atmospheric and low-pressure storage tanks constructed as per standards IS-803, API-620, API 650, IS 10987 or equivalent. The various types of storage tank inspections along with types of repairs and areas of inspections have been covered in this standard.

3.0 DEFINITIONS

3.1 Authorized Person

A qualified and experienced person authorized to perform storage tank inspections by the owner organization.

3.2 Shall

Indicates mandatory requirement.

3.3 Should

Indicates recommendation or that which is advised but not mandatory.

3.4 Corrosion Rate

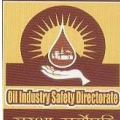
It is the total metal loss divided by the period of time over which the metal loss occurred.

3.5 Repair

Repair means any work necessary to maintain or restore a tank to a condition suitable for safe operation.

3.6 Applicable Standard

Applicable standard refers to the original standard of construction, unless the original standard of construction has been superseded or withdrawn from publication, in this event, applicable standard means the current edition of the appropriate standard.

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4.0 TYPES OF STORAGE TANKS & FLOATING ROOF SEALS

4.1 Atmospheric Pressure Storage Tanks

Atmospheric storage tanks are those tanks that have been designed to operate in its gas and vapour spaces at internal pressure approximately equal to atmospheric pressure.

4.2 Low Pressure Storage Tanks

Low-pressure storage tanks are those tanks, which are designed to operate at pressure in its gas or vapour spaces exceeding those permissible in API Std. 650, but not exceeding 1.06KG/SQ. CM gauge.

4.3 Fixed Roof Tanks

Among fixed roof tanks, cone roof tanks are very common for atmospheric storages tanks. Other fixed roof tanks are umbrella roof and dome roof tanks. Low-pressure roof tanks are generally constructed of dome roof.

4.4 Floating Roof Tanks

Floating roof tanks are designed to reduce filling and breathing losses to a minimum and for safety considerations by eliminating the vapour space above the stored liquid. There are mainly following types of floating roof tanks:

- i) Pan floating roof
- ii) Pontoon floating roof
- iii) Pontoon with Buoy type floating roof
- iv) Double Deck floating roof

The types of floating roof tanks and various seals are shown in Fig. 4.4.1, 4.4.2, 4.4.3 A and 4.4.3.B.

4.5 Fixed-cum-Floating Roof Tanks

Fixed-cum-floating roof tanks are fixed roof tanks with internal floating roof. These types of tanks are used for products, which are to be protected from contamination. These are also used at locations where snowfall is heavy.

4.6 Open Roof Tanks (Without Roof)

Open roof tanks are cylindrical vertical tanks with top open to the atmosphere. Roof is not provided and the material stored is exposed to the atmosphere. Open roof tanks are not used for storing hydrocarbons.

4.7 Horizontal Cylindrical Tanks

Horizontal cylindrical tanks are of two types;

(i) Above Ground Tanks

Above ground tanks are mounted horizontally above ground and are approachable externally.

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(ii) Under Ground Tanks

Underground tanks are placed in earth, masonry or concrete pit and packed around with sand, earth or clay leaving no air space between the tank and the pit.

4.8 Floating Roof Tank Seals

4.8.1 Primary Seal

The primary seals fall into three generic groups:

- I. **Mechanical Seals** – It consists of a metallic sealing ring or “shoe” held in permanent contact with the tank shell by a continuous sealing membrane bridging the gap between the top of the sealing ring and the pontoon rim angle to trap the product vapours.
- II. **Liquid Seals** – In this an elastomeric casing contains a quantity of compatible liquid, which “floats” on the product in the rim space to prevent evaporation.
- III. **Foam Seals** – In this case compressible foam is held in an elastomeric envelope, which traps the vapours at the top of the rim gap or prevents evaporation by resting on the stored product.

4.8.2 Secondary Seals

The secondary seals fall into two basic categories i.e. rim mounted and shoe mounted. The rim mounted secondary seals have the advantage since it replaces the need for weather-shields.

4.8.3 Double Seals

It consists of any primary seal mounted with any secondary seal. The most effective double sealing system is the liquid filled; product mounted primary Tube-seal with a secondary seal. Distorted tanks can be accommodated by provision of a tailor made extension to the rim angle.

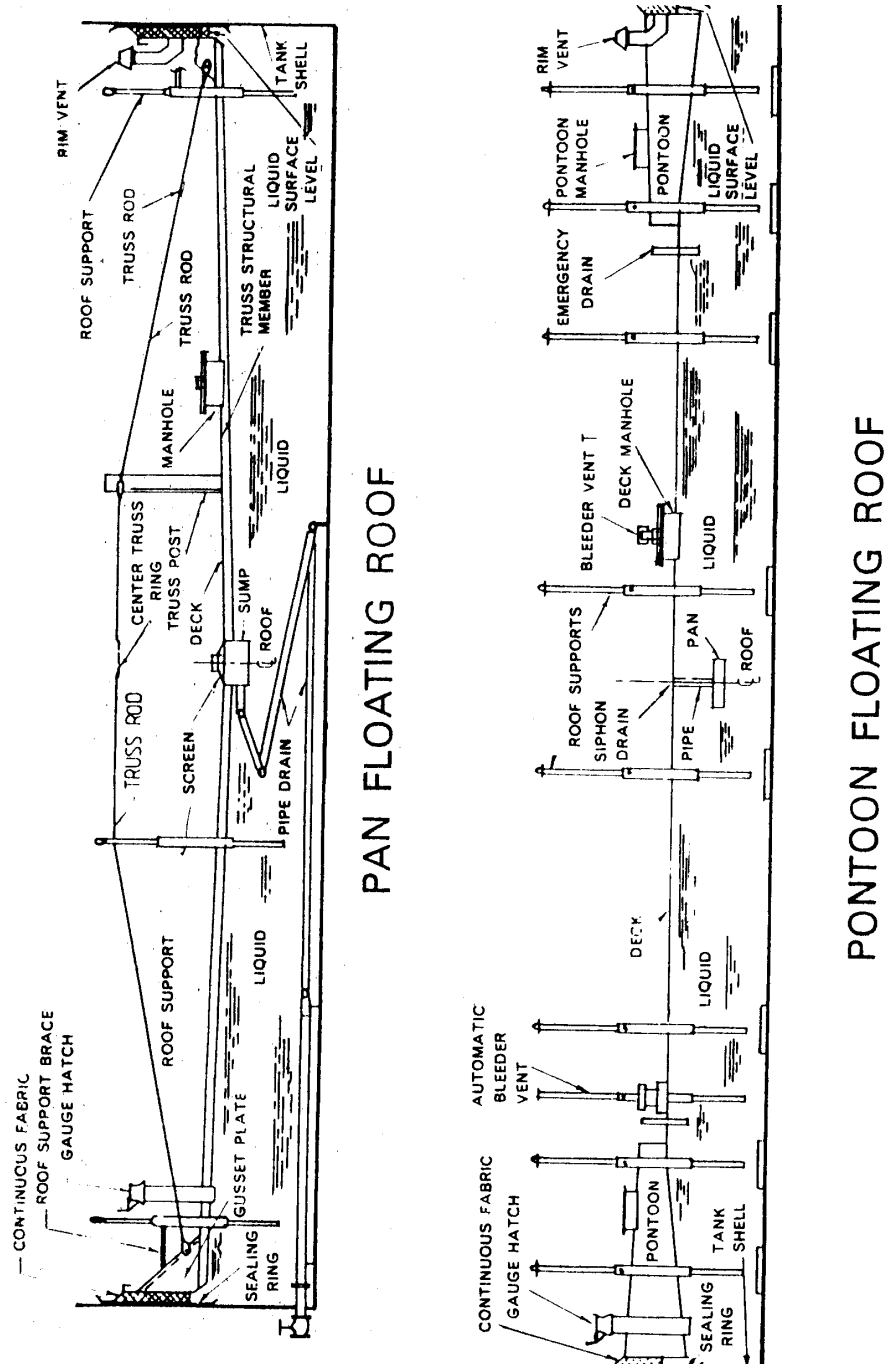
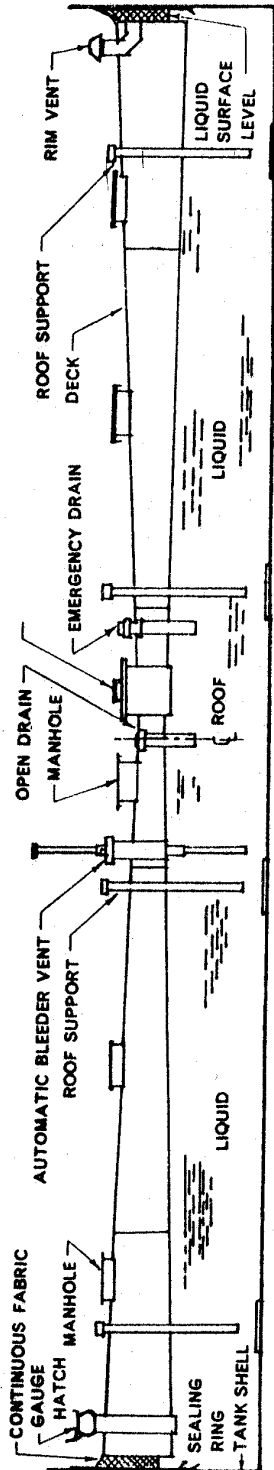
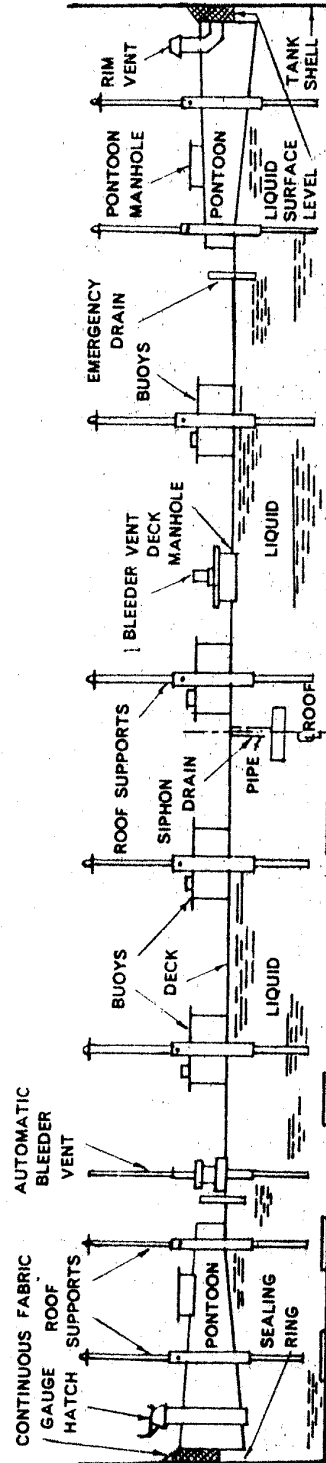


Fig 4.4.1 :CROSS-SECTION SKETCHES OF FLOATING ROOF TANKS SHOWING THE MOST IMPORTANT FEATURES.



DOUBLE DECK FLOATING ROOF



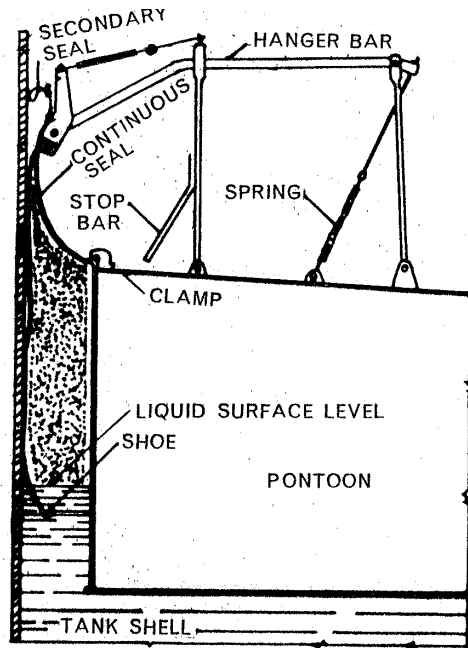
BUOY FLOATING ROOF

Fig 4.4.2 : CROSS-SECTION SKETCHES OF FLOATING ROOF TANKS SHOWING THE MOST IMPORTANT FEATURES

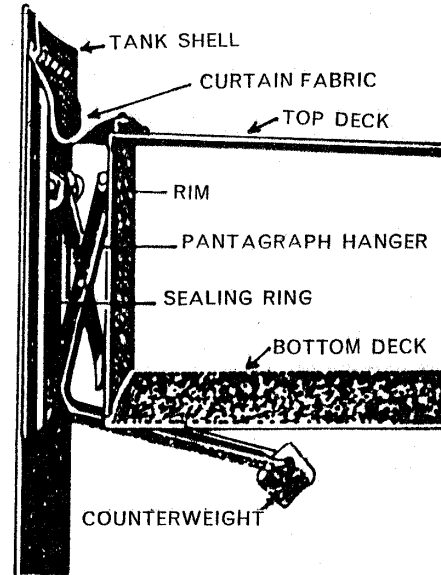


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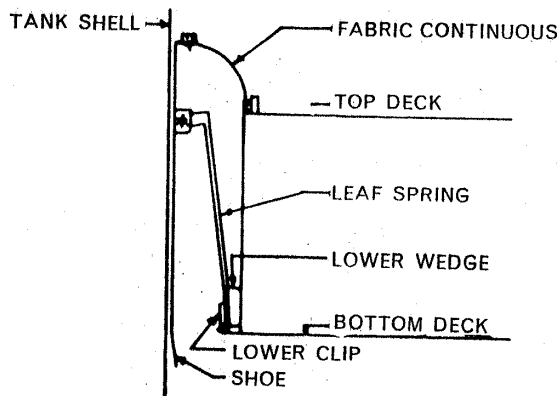
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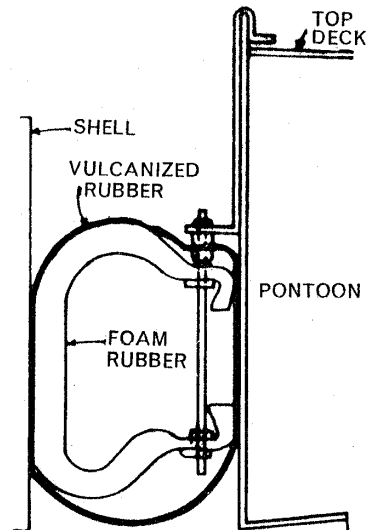
— Floating-Roof Seal Using Coil Springs to Maintain the Seal.



Floating-Roof Seal Using Counterweights to Maintain the Seal.

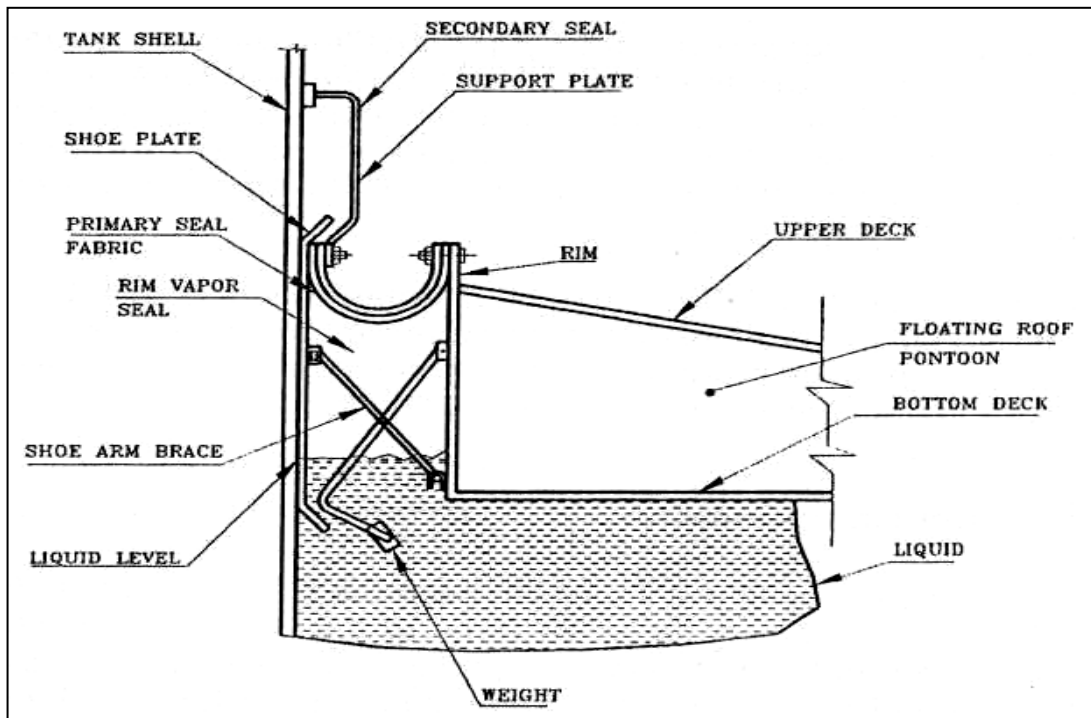


- Floating-Roof Seal Using Leaf Springs to Maintain the Seal.



- Floating Roof Using Resilient Tube-Type Seal.

Fig 4.4.3.A Floating Roof Seals



Mechanical Shoe Primary Seal with Rim Mounted Secondary Seal

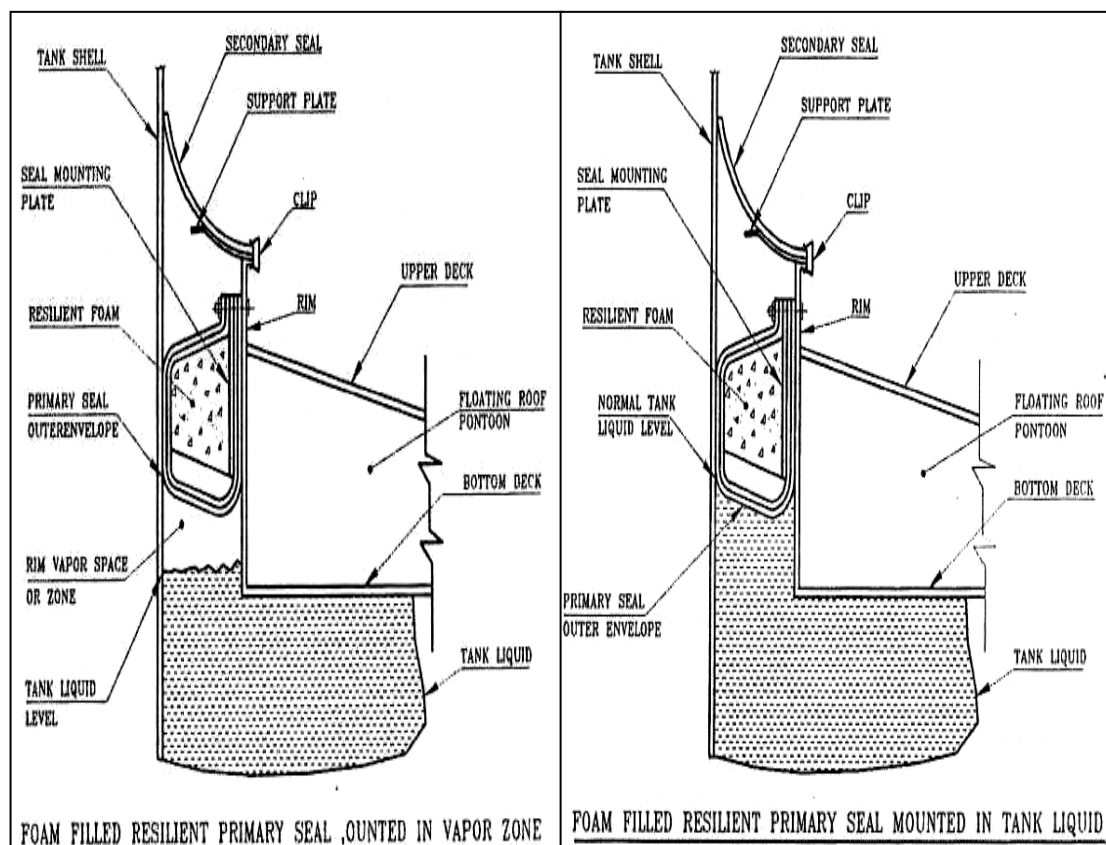


Fig 4.4.3.B Floating Roof Tanks Seals

from the use of OISD Standards/Guidelines."

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5.0 ROLE OF INSPECTION

The Authorized Person(s) performing the inspections shall be suitably qualified and experienced. The requisite criteria for deciding the qualification and experience shall be decided by the individual organization. Typical role of inspection personnel is;

- I) To prepare and implement tank inspection schedules to meet requisite standard, statutory and or quality requirements
- II) To measure and record the corrosion/ deterioration rates and to evaluate the current physical condition of the tank for soundness for continuation in service
- III) To co-relate the corrosion/ deterioration rate with design life for further run of the tank.
- IV) To investigate the causes of deterioration and recommend remedial measures, such as short term and long term repairs/ replacements.
- V) To perform various stages of inspections and maintain inspection records & tank history.

6.0 INSPECTION OF STORAGE TANKS DURING FABRICATION

Inspection of storage tanks during fabrication shall be carried out as per the requirements of the applicable codes, specifications, drawings etc. This inspection requires regular checks on the work at various stages as it progresses.

Following shall be ensured under this inspection;

- i) Study of all the technical specifications.
- ii) Checking the foundation pad and slope
- iii) Identification of plate materials
- iv) Qualification of welding procedure and welding operator
- v) Checking of underside painting of the bottom plate prior to its laying.
- vi) Checking of slope of the bottom plate.
- vii) Checking of each batch of electrodes as per specifications and assurance of its use as per recommended method by the manufacturer and codes.
- viii) Checking of proper welding sequence
- ix) Evaluating radiography of butt-welded annular (radial) joints and vacuum box test of the portion of weld on the bottom plate on which shell is to be erected.
- x) Checking of fit-ups and noting of curvature and plumb readings before and after welding of the shell courses.
- xi) Evaluating radiography of butt-welded joints as per the applicable code.

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- xii) A thorough visual check and oil penetrate test of the inside shell to bottom weld seam before welding from outside.
- xiii) Checking of nozzles/ man ways/ sumps for orientation, fit-ups and welding.
- xiv) Checking of set up of curb angle, roof trusses and roof plates prior to welding.
- xv) Checking of set ups and reinforcement arrangement of wind girders etc.
- xvi) Checking of PWHT of clean out doors, shell and shell nozzles, where applicable. After PWHT & before hydro testing, all such weld joints shall be inspected visually and tested using MT or PT.
- xvii) Checking of Nozzle pad for pneumatic test.
- xviii) Checking of external & Internal surfaces.
- xix) Perform the tests as specified below: -

A. For Fixed Roof Tanks

- a) Bottom plate test with vacuum box/air test.
- b) Hydrostatic Test as per Annexure-III.
- c) Roof air test or Roof vacuum box test.
- d) Rigidity/ Collapsibility Test (vacuum test)

B. For Floating Roof Tanks

- a) Bottom plate test with vacuum box/ Air test.
- b) Oil Penetrant Test of pontoon rims to bottom deck plate joint.
- c) Vacuum box test of top deck plates
- d) Pontoon air test/water test
- e) Roof drain hydraulic test
- f) Water fill-up test/Floatation test.
- g) Roof puncture test.

- xx) Steam Coil Hydrostatic Test
- xxi) Cooling system performance test
- xxii) Foam system performance test
- xxiii) Inspection of surface preparation and painting
- xxiv) Inspection of insulation wherever provided
- xxv) Stage-wise checking of settlement as per annexure-III.
- xxvi) Inspection of grounding connection
- xxvii) Checking of measures taken to avoid excessive distortion/ undulation of bottom and roof plates during fabrication.

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xxviii) Inspection for ensuring same level of sleeves of roof support pipes under the bottom deck of floating roof.

7.0 PRE-COMMISSIONING INSPECTION OF STORAGE TANKS

The pre-commissioning inspection of storage tank shall be performed to ensure that all examinations and tests during fabrication have been carried out. This inspection also includes the scrutiny of all the related records. A checklist for this inspection is placed at Annexure-V.

8.0 POST-COMMISSIONING INSPECTION OF STORAGE TANKS

8.1 GENERAL

The purpose of post-commissioning inspection is to assure continued tank integrity. It is necessary to draw up and adhere to an inspection programme to avoid failures and inconveniences in operation due to sudden reduction in tank storage capacity. The authorized person(s) shall carry out the internal and external inspections, except routine in-service inspections. The experience and qualification of the authorized person(s) shall be in line with the applicable inspection standards and procedures.

8.2 TYPES OF INSPECTIONS

- a) Routine In-Service Inspections
- b) External Inspection
- c) Internal Inspection

8.3 INSPECTION INTERVAL CONSIDERATIONS

Typical factors responsible for determining inspection intervals of storage tanks include the following:

- a) The nature of the fluid stored
- b) The results of routine visual/ external checks
- c) Corrosion allowances and corrosion rates/ trends
- d) Corrosion protective coatings
- e) Conditions at previous inspections
- f) The location of tanks such as isolated land, high-risk areas
- g) The potential of air & water pollution
- h) Corrosion prevention and leak detection systems
- i) Statutory regulations
- j) Industry's quality control requirements

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8.4 INSPECTION INTERVALS

The scheduling of post-commissioning inspections of a storage tank shall be as under;

a) Routine In-Service Inspection

The interval of such inspection shall be consistent with conditions at a particular site but site, but shall not exceed one month. Operations personnel, who have knowledge of the storage tank operations, shall carry out this inspection.

b) External Inspection

All storage tanks shall be given a Visual External Inspection at least once in a year by a qualified and experienced authorized person.

The Detailed External Inspection along with ultrasonic thickness survey of tanks shall be conducted as per Table No. 8.1. The detailed external inspection interval shall be determined using $D_T / 4N$, where D_T is the difference between measured shell thickness and the minimum required thickness in mm and N is the Tank Shell Corrosion Rate in mm per year.

S.N.	Fluid Stored	External Inspection Interval (Years)	
		With Corrosion Rate Based Assessment	Without Corrosion Rate Based Assessment
1	Crude Oil, Vacuum Gas Oil, Cycle Oil, SKO, MTO, ATF, HSD, Gas Oil, MS, Naphtha, Benzene, Toluene, Ethanol, MTBE, LDO, JBO, Bitumen, Lube Oil, Grease, Industrial Water, Amines	5	3
2	Fuel Oil, RCO, LSHS, Vacuum Residue, Slops, Caustic	3	3
3	Acids (Concentrate)	2	2
4	Acids (Dilute)	1	1

Table No. 8.1

A service history of a given tank or a tank of similar service, at the same location, should be maintained so as to schedule external inspection interval commensurate with the corrosion rate of the tank.

Illustration-

Fluid handled – Crude Oil

Let the measured shell thickness be 15 mm and designed minimum shell thickness be 3 mm.

Scenario 1; If measured shell corrosion rate is 1.0 mm per year, then.

$$D_T / 4N = (15 - 3) / (4 \times 1) = 3 \text{ years.}$$

Scenario 2; If measured shell corrosion rate is 0.5 mm per year, then

$$D_T / 4N = (15 - 3) / (4 \times 0.5) = 6 \text{ years}$$

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In the first case the tank shell shall be subjected to external thickness survey after 3 years and in the second case tank shell shall be subjected to external thickness survey after 5 years i.e. lesser of 5 years as per column (3) and 6 years as determined by calculation.

c) Internal Inspection

All storage tanks shall be subjected to a detailed internal inspection after an interval as detailed below in the **Table No. 8.2**. Where the inspection intervals are prescribed in the Quality Control Manual, the same shall override the table 8.2.

S.N.	Fluid Stored	Internal Inspection Interval (Years)	
		With Corrosion Based Assessment	Without Corrosion Based Assessment
1.	Crude Oil, Fuel Oil, RCO, LSHS, Vacuum Residue, Vacuum Gas Oil, Cycle Oil, SKO, MTO, ATF, HSD, Gas Oil, MS, Naphtha, Benzene, Toluene, Ethanol, MTBE, LDO, JBO, Bitumen, Lube Oil, Grease, Industrial Water, Caustic, Amines	Years determined by Corrosion Rate & Integrity Assessment or 15 years, whichever is lower	10
2	Slops	-Do-	8
3	Acids (Concentrate)	-Do-	4
4	Acids (Dilute)	-Do-	1

Table No. 8.2

The determination of internal inspection interval with corrosion based assessment shall require:-

- Establishment of Corrosion rate of the tank bottom plate based on previous inspection data of the same tank or anticipated based on experience with tanks in similar service at the same location.
- Assessment of tank bottom plate integrity through the use of on-stream Non Destructive Evaluation (NDE) methods such as Acoustic Emission, Guided Wave Ultrasonic Thickness Measurement, Robotic Measurement etc.

It must be ensured that the bottom plate thickness at the next internal inspection is not less than the minimum/ retiring thickness.

In no case, the internal inspection interval shall exceed 15 years.

The inspection observations, records pertaining to the corrosion rate calculation and integrity assessment of the tank shall be retained in auditable format.

Illustration

Fluid handled – Crude oil

Let the measured bottom plate thickness be 8 mm and designed minimum/ retiring bottom plate thickness is 3 mm.

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Scenario 1; If measured bottom plate corrosion rate is 0.4 mm per year, then.
Next Internal Inspection Interval = $(8 - 3)/0.4 = 12.5$ years.

Scenario 2; If measured bottom plate corrosion rate is 0.25 mm per year, then
Next Internal Inspection Interval = $(8 - 3) / 0.25 = 20$ years

Scenario 3; If measured bottom plate corrosion rate is 1.0 mm per year, then
Next Internal Inspection Interval = $(8 - 3) / 1.0 = 5$ years

In the first case the tank Internal inspection shall be done after 12.5 years based on corrosion assessment, in the second case tank Internal inspection shall be done after 15 years i.e. lesser of 15 years fixed as maximum or 20 years as determined by calculation and in the third case the tank internal inspection shall be done after 5 years based on corrosion assessment.

8.5 INSPECTION CHECKS

The inspection checks involved for different types of inspections are detailed below;

8.5.1 Routine In-Service Inspection

The external condition of the tank shall be monitored by close visual inspection from the ground on a routine basis. Personnel performing this inspection should be knowledgeable of the storage facility operations, the tank, and the characteristics of the product stored.

The routine in-service inspection shall include a visual inspection of the tank's exterior surfaces. Evidence of leaks; shell distortions; signs of settlement; corrosion; presence & growth of vegetation on pavement and condition of the foundation, bund wall, paint coatings, insulation and appurtenances should be documented for follow up action.

8.5.2 External Inspection

The storage tank shall be subjected to annual visual inspection checks and detailed external inspection at an interval as detailed above in 8.4.b.

8.5.2.1 Annual Visual Inspection checks

As part of the external annual visual inspection of a storage tank by a qualified & experienced authorized person, the following checks, in addition to the routine in-service checks, shall be performed;

a) Protective Coating

Condition of paint/ protective coating shall be checked visually for any deterioration such as rust spots, mechanical damage, blisters and film lifting etc.

b) Roof Plates

Roof plates shall be inspected for defects like pinning holes, welding cracks, pitting etc. Roof shall be checked for water accumulation and product flow marks on the roof due to leaks. Check for any sign of tilting of floating roof.

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c) Ladders, Stairways, Platforms and Structures

These shall be inspected for corrosion or and broken parts. Free movement and alignment of wheels of rolling ladder on rails shall be checked. In addition to loss of strength caused by loss of metal, staircase steps become slippery when the surface is worn. Handrails shall be checked for firmness. Platforms and walkways shall be inspected for thinning, water accumulation areas and general corrosion.

d) Tank Pad

- i) Tanks pads shall be visually checked for settlement, sinking, tilting, spalling, cracking, grass/ weed growth and general deterioration. Attention may be given to the area where water from the tank cooling/ FW system nozzle leaks which may damage the tank protective layer. Additionally, oil leak from the mixers of the tanks may soften the bitumen carpet layer there by loosing the strength to hold sand. Such loosened bitumen carpet layer shall be repaired.
- ii) Proper sealing of opening between tank bottom and the pad shall be checked (no water shall flow under the tank). Due to settlement of the tank, the exposed portion of the annular plate may get covered with bitumen/ cement, which shall be corrected.
- iii) Slope of tank pad shall be checked to ensure water drainage.

e) Anchor Bolts

Anchor bolts wherever provided shall be checked for tightness, and integrity. These shall also be checked for thinning/ bending. Distortion of bolts is an indication of excessive settlement. Concrete foundation at anchor bolts shall be checked for cracks.

f) Fire Fighting System

Condition of the firefighting system shall also be checked for general corrosion General condition of firefighting facilities and tank cooling sprinkler system provided for the tank shall be checked for general condition with respect to clogging of spray nozzles, performance of foam connections, etc, Frequency and procedure for checking shall be as per OISD-STD-142 (Inspection of Fire Fighting Equipment).

g) Vents & Pressure Relieving Devices

All open vents, flame arrestors and breather valves shall be examined to ensure that the wire mesh and screens are neither torn nor clogged by foreign matter or insects. Rim vent and bleeder vents for floating roof tanks shall be examined for proper working. All vents and pressure relieving devices shall be inspected as per the frequency and procedure outlined in OISD-STD-132 (Inspection of Pressure Relieving Devices).

h) Insulation

The insulation, weatherproof sealing and straps around the insulation cladding shall be inspected for damage. The waterproof sealing of the insulation shall be examined. At locations where external corrosion is predominant due to saline atmosphere/ heavy rain, the insulation from tank bottom shell course approx. 200 mm shall be kept bare to avoid accumulation of soaked water and corrosion of plate. The inspection pockets shall be kept closed. In case of fixed roof tanks, the roof insulation cladding shall be kept little extended beyond the curb angle.

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i) Grounding/ Earthing Connections

Tank grounding/ earthing system components, such as roof guide roller/ shunts or mechanical connections of cables shall be checked visually for corrosion or any other damage. Grounding connection shall be visually checked for corrosion at the points where they enter earth and at the connection to the tank. The resistance of grounding connections shall be checked annually before monsoon. The resistance from tank to earth shall not exceed 7 Ohms and the resistance from any part of the fitting to the earth plate or to any other part of fitting shall not exceed 2 Ohms.

j) Leaks

The tanks shall be inspected for any obvious leakage of the product. Valves and fittings shall be checked for tightness and free operations. Tank pads shall be checked for wetness and softening due to leaks from tank bottom plate. Check for passing/ leaking of floating roof drains. Any evidence of presence of oily substance or water in pontoon boxes should be examined.

k) Tank Mountings

Tank mountings such as Breather Valves/ P&V Valves, Relief Valves, Flame Arrestors etc. shall be ensured clean and operable after monsoon. Floating roof drains, shell sealing and emergency roof drains shall be inspected before monsoon.

8.5.2.2 Detailed External Inspection Checks

The detailed external inspection checks shall also include visual inspection checks as mentioned above. The tank may be in operation during this inspection. The inspections shall be done using safe working practices and following safety measures specified at Annexure-II.

Prior to inspection of the tank, the as built drawings, history card and previous inspection shall be referred. An inspection checklist shall be prepared and kept ready. A typical checklist for external inspection is placed at Annexure- I (a)

The following shall be inspected / checked during this inspection.

a) Tank fittings, Accessories and Pipe Connections

All nozzles shall be inspected for corrosion/ distortion. Reinforcement pads, if provided, shall be checked for evidence of leak through the telltale holes. All nozzle pipes shall be checked for thickness. On nozzles of size 50 mm NB and above, minimum 4 readings (3,6,9 & 12 O'clock positions or on East, West, North and South direction) shall be taken. For small-bore nozzle pipes, the inspector shall determine extent of such measurements.

b) Tank Shell

The tanks shell shall be visually examined for external corrosion, seepage, cracks, bulging/ buckling and deviation from the vertical. Wind girder condition shall be assessed including wind girder plates and adjacent shell wall, stitch welds of girder with stiffener plates, butt welds of wind girder, weather deflector plate etc. The condition of external painting shall also be checked for any deterioration.

External thickness measurement of the shell can be a means of determining a rate of uniform general corrosion while the tank is in service and can provide an indication of the integrity of the shell. External thickness survey shall be carried out all around the bottom shell course. For the balance shell courses, thickness survey shall be carried out as outlined below. An extensive

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scanning shall be done if there is an indication of appreciable metal loss. The likely areas of metal corrosion/ wastage are detailed in Annexure-VI.

The following minimum requirement for thickness survey is recommended on all the tanks:

- i) All the plates of bottom two-shell course should be checked for thickness.
- ii) On the first course, the readings shall be taken in such a way that the bottom, middle and top positions of each plate are checked for thickness. An average of a minimum of 4 readings shall be taken on each plate diagonally to arrive at the remaining thickness.
- iii) On the second course, thickness measurement shall be carried out at two elevations to cover all plates. One location shall be very close to the bottom weld joint and the other at the approachable height. An average of a minimum of 4 readings shall be taken at each location to arrive at the remaining thickness
- iv) For the balance shell courses, thickness measurement shall be taken at three elevations covering bottom, middle and top of the shell plate approachable from the spiral staircase. An average of a minimum of 4 readings shall be taken at each plate to arrive at the remaining thickness.

For tanks in lighter products service such as Motor Spirit and Naphtha, pitting is generally observed in the middle courses of the shell due to frequent wetting and drying of the shell plates at this elevation. In such cases, thickness survey should be more extensive on middle shell courses. In addition, these shell plates in a floating roof tanks shall be checked visually for sign of corrosion and pitting. If, due to some reason, significant internal corrosion of roof is observed, then top shell course(s) should also be examined for thickness. In case of externally insulated tanks, suitable inspection windows shall be provided to facilitate wall thickness survey. These inspection windows shall be closed using proper covers during normal operation to ensure that water does not enter insulation.

For the tanks which are likely to have water at the bottom, the bottom shell courses near the annular ring weld joint should be thoroughly checked ultrasonically for 150mm of the bottom plates.

For determining the limiting thickness for the shell plates of a tank, either for the purpose of pre-calculating a set of retiring thickness for each tank or as a matter of necessity at the time of inspection, the basic method given in the applicable standard shall be used. Further details on calculation of rejection thickness of shell plates are given at Annexure VII.

c) Tanks Roof

i) Fixed Roof

Visual inspection of the roof shall be made to determine condition of paint and to check for depressions, sagging and perforation/ holes, if any. Ultrasonic thickness survey shall be carried out to cover all roof plates.

Insulated tanks shall be externally inspected for detecting corrosion by removing inspection covers or making pockets in insulation. These inspection windows shall be closed using proper covers during normal operation to ensure that water does not enter insulation.

If a tank is out of service, hammer testing along with the thickness survey of the roof plates shall be carried out. Insulation on the roof shall be checked for wetness and damages to the cladding.

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Cladding shall be sealed properly to avoid water ingress. Any tear/ damage to the cladding plates and sealing shall be repaired

On fixed roof tanks, planks/ grating, long enough to cover at least two roof rafters as bottom support shall be laid and used as walkways for safety reasons. Corroded roof plates may be a safety hazard for people to walk.

ii) Floating Roof

The following shall be checked on a floating roof during inspection:

a) Condition of painting / protective coating

b) Undulations/ depressions on the deck plate where water can accumulate. In addition to water accumulations, these act as pockets of vapours below the roof. One should be very careful while walking on the roof as these vapours gets released while walking on these spots through the nearest support openings / vents available

c) Pontoon boxes and buoys for leakages/ seepages and corrosion.

d) Roof drains and emergency roof drain:

Drain shall be checked for breakage and blockage on the check valve fitted to the roof drain inlet end. The debris hinders the free movement of the check valve flapper and may also block the drain piping inside. Emergency drains shall be checked for water seal level and oil spill mark on roof. Both the drains (primary and secondary roof drains) shall be checked every year before monsoon. Welds of check valve nozzle shall be checked for any deterioration or crack. LPT shall also be carried out.

e) Dip hatch / anti rotational device may be checked for rubbing against the pontoon box top through which it passes. Ultrasonic thickness measurement of Dip Hatch pipe should be taken and recorded. The plumbness of the pipe should also be checked.

f) Cable connection between the roof and the rolling ladder and ladder to tank shell for electrical continuity shall be checked. Check for the cleats where it is connected for corrosion and mechanical damage.

g) Floating roof seals shall be inspected visually. Seal assembly shall be inspected for corrosion, erosion, tears, broken parts and deteriorated sealing materials. Exposed mechanical parts such as springs, hangers, counter-balance, pantographs, shoes and weather protectors are susceptible to mechanical damage, in addition to wear and atmospheric/ vapour space corrosion. Ensure that the rubber seals shall have a fairly close contact with the tank shell plates. In case of liquid filled seals, check for any indication of leakage.

h) Condition of hinge and connecting bolts at the top of ladder and near the rollers shall be checked for wear and corrosion.

i) Roof shall be observed for free movement without any obstruction both in lateral and rotational direction and for tilting

j) Weld seams on the deck plate for any evidence of leakage

k) Condition of roof support legs, (both pipes and sleeves) and sealing cap if provided between the support pipe and sleeve shall be inspected for corrosion / damages

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l) DP Test of bottom deck weld at overhead portion, plumbness of legs and LP test of welds on legs shall be carried out.

d) Projecting Out Portion Of Bottom Plates

The projecting out portion of the bottom plate (generally annular plate) shall be inspected for corrosion, thinning (using Ultrasonic Thickness Measurement) and damages. Liquid Penetrate Test (LPT) of shell to annular plate shall be carried out after mechanical cleaning preferably grit blasting. Ensure that this portion shall not be covered with soil/ bitumen/ cement due to tank settlement

8.5.3 INTERNAL INSPECTION

Internal inspection of storage tanks is conducted for the following reasons;

- i. Check for the bottom plate condition for corrosion and leaks.
- ii. Gather the data necessary for minimum thickness assessments to arrive at corrosion rates.
- iii. Identify and evaluate tank bottom in-service settlement.
- iv. Inspection and health assessment of internal heating coils.
- v. Ensure condition of crude oil side entry mixers.
- vi. Ensure condition of roof drain assemblies.
- vii. Ensure health of roof supporting structures.

All tanks shall have a formal internal inspection conducted at the intervals as detailed in item 8.4.(c). The internal inspection shall be carried out by qualified & experienced authorized person/s. The tank must be prepared for man entry by emptying of its liquid, gas freeing, cleaning A typical checklist for internal inspection is placed at Annexure- I(b)

It shall be ensured that the tank has been thoroughly cleaned off lead hazards and is safe for man entry. The inspections shall be done using safe working practices and following safety measures specified at Annexure-II.

The internal inspection of a tank shall also include an external inspection of the tank as outlined in section 8.4.b. A typical list of tools required for tank inspections is placed at Annexure-IV.

The following checks shall be performed during internal inspections;

a) Roof and Structural Members

i) Fixed Roof

Visual inspection and thickness measurements of roof trusses and structural members shall be carried out. The supporting members shall be rejected when the overall loss in thickness of material exceeds 25 percent of initial thickness. In addition, all structures may be hammer tested. The welds and bolts of the structure shall be examined for damage. The damaged bolts should be rejected and replaced. The results of measurements on the supporting structure may be recorded as shown in Fig.8.5.3.1

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ii) Floating Roof

The underside and internals of floating roof shall be inspected for corrosion and deterioration. The floating roof seals shall be inspected from the underside for tears / damages. The leg supports and sleeves of the floating roof shall be checked for deterioration, bowing, and shifting. The dip pipe, centering and anti-rotational devices, emergency roof drain pipe, free/ breather vents, rim vents etc. shall be checked for sign of corrosion/ thinning out. The weld joints of Rims, stiffeners and deck plate shall be inspected. Chalk oil test should be conducted on rim welds and bottom deck.

Vapour space below the roof and the portion of the roof deck plate near the undulation shall be checked for signs of corrosion / pitting. Thickness survey of the deck plates, pontoon box plates shall be carried out. Any suspected pontoon box/ buoy may be checked for integrity using oil chalk or pressure test.

b) Tank Shell

Entire tank shell shall be visually scanned for signs of corrosion, pitting, cracking, deterioration etc. Findings of external inspection, service condition and tank history will be guiding factors for such observations. All weld joints shall be examined carefully for cracks, grooving or mechanical damages. Vapour space and liquid level line are likely areas of corrosion. Since walls are alternatively wet or dry and if the product is corrosive, the entire shell is prone to corrosion. When severe corrosion is seen on the shell at various locations especially in Motor Spirit or Naphtha tanks, it is necessary to build approach and inspect entire shell length.

c) Tank Bottom

After the tank has been cleaned off of the sludge, the plates shall be visually inspected to get the first indication of the condition of the bottom. The likely areas of metal corrosion/ wastage are detailed in Annexure-VI.

The tank bottom plates shall be visually inspected for pitting, corrosion and weld cracks. The weld joints shall be thoroughly cleaned and visually inspected for cracks. Suspected cracks may be further checked by Penetrant Test (PT) or Vacuum Box Test or Magnetic Particle Test (MT). Depressions in the bottom plates, bottom plates under the roof supports and area below heating coil supports shall be checked closely. Any water getting into the tank may collect and remain at these points thereby causing accelerated corrosion. Tank bottom shall be checked thoroughly for thickness over the entire area. This may be supplemented by hammer testing. The number of measurements to be taken will depend on the size of the tank and the degree of corrosion found. However, minimum three thickness readings per plate should be taken. When severe corrosion exists, more readings shall be taken in the corroded areas to determine the minimum metal thickness in that area. As an alternative, a suitable non-destructive testing method, which scans the entire bottom plate and gives the profile of the entire bottom plate, can also be used. Techniques such as Magnetic Flux Leakage (MFL) or Low Frequency Electromagnetic Testing (LFET) coupled with ultrasonic thickness measurement can be used. Suitable care shall be taken to prepare the tank bottom to achieve the level of cleaning required for techniques like MFL / LFET.

Corrosion of the bottom plates on the soil side of the flat bottom tanks is not visible from oil side of the tank. Erratic readings with ultrasonic thickness measurement are indications of soil side corrosion. To carry out a positive inspection and accurate check, it is recommended to cut out representative sections of coupons (at least 300 mm in least dimension) of the bottom plate. The underside of the coupons shall be inspected. Additional coupons shall be removed from tank bottom plate, If very severe corrosion is found. A suitable NDT, which maps the bottom

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plate thickness, taking into account both oil side as well as soil side corrosion, can be used as an alternative to removal of test coupons. The cut out opening in the bottom plate shall be patch welded using new plates of appropriate thickness and size. The welds of the patch plate shall be tested with vacuum box or Magnetic Particle Testing (MT). If tank is suspected leaking, then the cutting operations shall be done under strict observations because of possible entrapped hydrocarbons.

The tanks with water bottoms (such as crude tanks) shall be given special attention to check for corrosion of the bottom plates. The bottom plates where bacterial corrosion may be suspected (such as crude and HSD tanks), the area shall be gauged in more detail and shall be checked for corrosion under deposits.

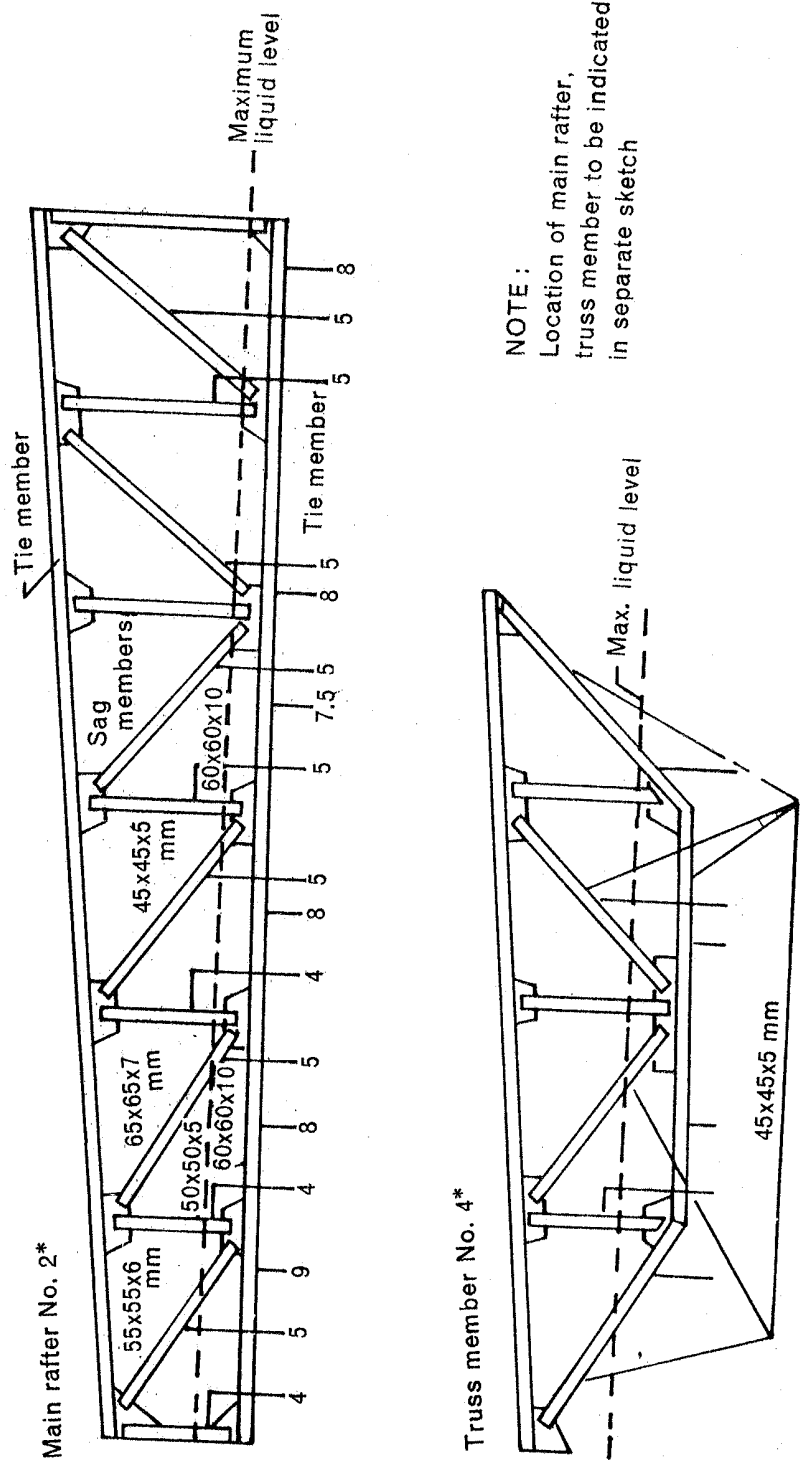


Fig. 8.5.3.1 TYPICAL EXAMPLE OF ROOF-STRUCTURE MEASUREMENTS

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d) Water Draw-off

Water draw off are subject to internal and external corrosion as well as cracking. They shall be visually inspected and hammer tested along with thickness measurement. Bottom plates below dip hatch shall be checked for dents, cracks, corrosion and mechanical damage.

Drain sumps shall be carefully checked for cracks, pitting, leak in the weld, and measured in particular when corrosion of the underside of the bottom plate is suspected/ found.

e) Linings/ Painting

When the inside surface of a tank is coated or lined with corrosion resistant material such as sheet lead, rubber, organic and or inorganic coating or concrete, inspection shall be made to ensure that the lining is in good condition and position without any cracks / holes / discontinuities / bulging etc, Care must be used to ensure that the voltage used during holiday testing does not puncture the lining.

If the tank is rubber-lined hardness testing of the rubber lining shall be checked. Care shall be taken while cleaning the painted surface against mechanical damages

It is recommended to ascertain the remaining life of internal linings / internal painting provided in tanks. The lining / painting provided should have adequate remaining life to last till next inspection interval.

f) Roof Drains

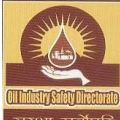
Roof drains on the floating roof tanks can be designed in many ways. They can be simple open drainpipes, swivel joints or flexible hose drains that keep the water from contaminating the contents. Proper functioning of the roof drains shall be ensured otherwise this may lead to tilting/ sinking/ over-turning of the floating roof. The drain lines shall be checked for blockages prior to pressure test. All swivel joints shall be thickness surveyed and serviced during every outage and individually hydro tested. After assembly of the roof drains system, complete system shall be hydro tested. Roof drain manufacturers recommendation for inspection, servicing and testing of roof drain system shall be followed. If a check valve is fixed on the tank roof at the roof drain opening, the same shall be serviced and tested. The weld joints on check valve nozzle shall be checked by LPT and if found cracked reinforcements should be carried out.

Emergency roof drain sump shall be checked for leaks. Emergency drainpipe shall be checked for corrosion, thinning and cracks. Emergency roof drain sump shall be filled with clean water before boxing up of the tank.

g) Heating Coils

Heating coil including the supports shall be hammer tested, particularly at the underside of coil and bends. Ultrasonic thickness measurement shall be taken.

Random radiography of bends and piping section shall be taken for evaluation, if internal corrosion is suspected. The heating coil shall be thoroughly inspected for corrosion, erosion, cracks, etc. The slope of the heating coils shall be checked for proper draining of condensate. Sample pipe piece can be cut & removed for sectioning and assessment of internal corrosion. Coils shall be hydraulically tested at 1.5 times operating pressure and checked for any leaks.

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h) Miscellaneous

All valves, other mountings and fittings shall be checked for leakage and proper functioning. All valves including breather valves shall be serviced and reset at the required pressure and vacuum settings.

9.0 REPAIRS AND INSPECTION

Methods described are recommendatory in nature. Other methods conforming to sound engineering practice may also be applied. These repair methods have been outlined to highlight the inspection required prior to, during and after repairs. Unless otherwise stated, the requirements and acceptance criteria as per the original construction shall prevail. The details of repairs and inspection of storage tanks are enclosed at Annexure VIII

10.0 INSPECTION OF UNDERGROUND STORAGE TANK

Prior to entering an underground tank, it shall be cleaned internally of its product and adequate air circulation provided. The man entry should be preceded by ensuring that the tank is Hydrocarbon & gas free. The tank shall be visually inspected for general corrosion/ pitting/ deterioration on internal surfaces.

Ultrasonic thickness measurements shall be carried out on shell plates, end plates and nozzles from inside the tank once in ten-year period or to meet industry's quality control requirements whichever is less. The external inspection of the underground tank shall be performed once in twenty-year period or to meet industry's quality control requirements whichever is less..

After the repairs, the tank shall be hydrostatically tested of at 0.75-kg/sq.cm-test pressure and checked for leaks.

11.0 DOCUMENTATION

11.1 DOCUMENTATION FOR NEW TANKS

The following completion documents for the storages tank shall be preserved:

- i) As built drawing of the tank.
- ii) As built drawing of the tank foundation.
- iii) Shell development drawings indicating the location of radiography and test results of the shell.
- iv) Radiography films (for 5 years after completion)
- v) Bottom layout and test results of the bottom.
- vi) Heating coil layout and test results.
- vii) Certificate of earthing of a tank.
- viii) Roof layout and test results.
- ix) Settlement results of tank bottom.
- x) Pontoon and floating roof test results.

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- xi) Test certificates of seal.
- xii) Calibration charts.
- xiii) Certificate of tank mountings.
- xiv) Floating roof drainage system drawing and test results.
- xv) General arrangement drawings with design data and material specification.
- xvi) Nozzle orientation drawing.
- xvii) Stairway details with orientation.
- xviii) Foam system drawing.
- xix) Cooling system drawing
- xx) Wind girder drawings
- xxi) Pipe support drawings with orientations
- xxii) Rolling ladder detail drawing.

11.2 DOCUMENTATION FOR TANKS IN-SERVICE

All the observations, findings and repairs carried out after each inspection should be recorded. As a minimum, the data given in the typical formats given below should be recorded;

- a) Tank inspection and repairs card (Ref. Fig. 11.2.1)
- b) Data Record History Card (Ref. Fig. 11.2.2)

The corrosion rate calculations and integrity assessment observations shall be recorded in formats.

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Fig 11.2.1

TANK INSPECTION AND REPAIRS CARD

REPORT ON TANK INSPECTION AND REPAIRS		
Inspected on:	Tank No.:	
Location :		
Dimensions :		
Type of Tank :	Product:	
Year of Construction : 19		
Previous Internal Inspection :		
Previous external Inspection :		
Type of inspection : Internal/external		
Summary:		
	Inspection Observations	Repairs Executed
Foundation :		
Bottom :		
Shell :		
Roof Structure :		

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Fig 11.2.1(cont)

	Inspection Observations	Repairs Executed
Roof : Fixed/Floating Pontoon Ladder hinges		
Miscellaneous Suction and inlet devices Free vents Breather valve Splash plate Spray water riser Foamite spray riser Foam connections Dip hatches Wind shield of dip hatch Roof manhole Shell manhole Liquid level indicator Hand railing Stairways Product drain Water draw-off Measuring point Nozzles Mixing coils Heating coils Shunts Hinge bolts and roller		
Insulation Painting		
Recommended date for next inspection: Repairs expected during next inspection:		
Remarks :		
Inspected by 	(Name)	Date (Signature)

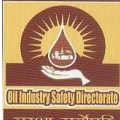


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Fig Fig 11.2.2

TANK DATA RECORD CARD				SHELL DETAILS		APPURTENANCES	
INFORMATION							
Manufacturer _____ Manufacturer's Drgs _____ Reference Drg. _____ Order Number _____ Type _____ Code _____ Size : Diameter _____ Height _____ Capacity _____ In use since _____				Horizontal seams Vertical seams Number _____ Matl. _____ Thk. _____		External P & V Valves _____ Nozzles & Pipe Connections _____ Nozzle No. _____ Pipe _____ Flange _____	
DESIGN/OPERATING CONDITIONS Service _____ Location _____ Temp. °C Design _____ Operating _____ Press Kg/cm² Design _____ Operating _____				BASE DETAILS Radiographed _____ Sketch Plates _____ Thickness _____ Joints _____ Annular Plates _____ Thickness _____ Joints _____ Type of foundations : _____		INTERNAL Swing Pipe & Accessories _____ Heating Coils _____ Mixing Nozzles _____ Float & Guide _____ Main Inlet Extension _____ Drain Sump _____ Guide Section _____ Flexible Drain Pipe _____ Roof Supports _____	
ROOF DETAILS Part _____ Matl. _____ Thk. _____ Roof Plates _____ Crown Plates _____ Deck Plates _____ Supports _____ Size _____ Rafters _____ Inter Rafters _____ Tertiary Rafters _____ Purlins _____ Verticals _____ Diagonals _____ Ties _____ Bracings _____ Top Crown Drum _____ Top angle (Curb) _____ Pontoons _____				PAINTING Internal Roof _____ Trusses _____ Shell _____ Bottom _____ External Primer _____ Colour _____			

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12.0 REFERENCES

The following codes, standards and publications have either been referred or used in the preparation of this standard and the same shall be read in conjunction with this standard.

- i) API RP 575 Guidelines and methods for inspection of existing Atmospheric and low pressure storage tanks
- ii) API 650 - Welded Steel tanks for oil storage.
- iii) API 620 - Design and construction of large welded low-pressure storage tanks.
- iv) BS EN 14015:2004 Specification for the design and manufacture of site built, vertical, cylindrical, flat bottomed, above ground, welded, steel tanks for the storage of liquids at ambient temperature and above.
- v) IS 803 - Code of practice for Design, Fabrication and Erection of Vertical M.S. Cylinder Oil storage tanks.
- vi) IS 4682 - Code of practice for Lining of Vessels and Equipment for Chemical Processes. Part - 1 Rubber lining
- vii) API STD 2015 – Safe Entry and Cleaning Petroleum Storage Tanks
- viii) API Publ 2026 – Safe access/ Egress involving floating roof of storage tanks in petroleum service
- ix) API RP 2003 - Protection against ignitions arising out of Static, Lighting and Stray Currents
- x) IS 9964 - Recommendations for Maintenance and Operation of Petroleum Storage
 - Part -I -- Preparation of tanks for safe entry & Work.
 - Part –II -- Inspection.
- xi) IS 10987 - Code of practice for Design, Fabrication Testing and installation of underground / aboveground cylindrical storage tanks for petroleum products.
- xii) API 2000 - Venting Atmospheric and Low-Pressure Storage tanks.
- xiii) API 653 – Tank Inspection, repair, alteration and reconstruction

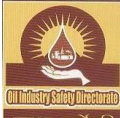
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Annexure-I (a)**TYPICAL CHECKLIST FOR EXTERNAL TANK INSPECTION****LOCATION :** _____**TANK NO.:** _____

Capacity (KL) :	Dia (Mt) :	Height (Mt) :	Type of Tank :
Construction Year :	SFH :	No. of Courses :	Product :
Last Inspection Date :		Current Inspection Date :	

EXTERNAL INSPECTION CHECKS

1.0	Tank Pad Foundation	Observation	Recommendation
1.1	Condition of Foundation pad		
1.2	Circumferential drain is free from water stagnation, debris etc.		
1.3	No voids found between annular plate extensions and foundation surface.		
1.4	Foundation slope is satisfactory to avoid water stagnation		
1.5	Abnormal/ uneven settlement.		
2.0	Annular Plate Extension		
2.1	Annular plate extensions are free from corrosion, pitting, scales and the painting condition is satisfactory.		
2.2	Annular plate thickness		
3.0	Condition of Shell Plates		
3.1	Corrosion and the painting condition		
3.2	Abnormal dents / buckles.		
3.3	Sweating / leaks /cold weld /repairs /patch plates		
3.4	Reinforcement pads of nozzles.		
3.5	Supports for connected Pipelines		
3.6	Wind girders		
3.7	Verticality of the tank		
3.8	Thickness readings at nozzles for inlet/outlet pipes, manholes, clean outdoors etc.		
3.9	Shell plate's thickness readings		

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4.0	Condition of Fixed Roof Plates		
4.1	Corrosion and painting		
4.2	Roof plates weld joints		
4.3	Patch plates / cold weld repairs.		
4.4	Roof sagging / stagnated water / water marks.		
4.5	Curb angle and its weld joint condition with shell plates		
4.6	Thickness readings of roof plates and Nozzles		
5.0	Condition of Floating Roof		
5.1	Corrosion and painting		
5.2	Roof plates weld joints		
5.3	Roof patch plates / cold weld repairs.		
5.4	Roof sagging / stagnated water/water marks.		
5.5	Thickness readings of roof plates, nozzles		
5.6	Pontoon and painting		
5.7	Primary Seals		
5.8	Deck support legs and pipe sleeves		
5.9	Main articulated drain / emergency drains along with water seal pots		
5.10	Rim vents / PV valves		
5.11	Auto bleeder vents		
5.12	Sample/dip hatch pipe and rollers/ brass sleeves		
5.13	Rolling ladder		
5.14	Gap between tank shell and roof rim		
5.15	Weather shields / secondary seal		
5.16	Condition of Foam dam		
5.17	Gap between gauge pipe and sleeve pipe		
5.18	Earthing connections between roof to ladder & ladder to shell plate		
6.0	Condition of Appurtenances/Attachments		
6.1	Free vents, Valves/ Flame Arrestors		
6.2	Earthing connections		
6.3	Earthing pits of the tank		
6.4	Insulation, inspection pockets and claddings		
6.5	Sealing of insulation around man-ways , nozzles and curb angles		
6.6	Staircase, handrails etc. and paintings		

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Annexure-I (b)**TYPICAL CHECKLIST FOR INTERNAL TANK INSPECTION****LOCATION :** _____**TANK NO.:** _____

Capacity (KL) :	Dia (Mt) :	Height (Mt) :	Type of Tank :
Construction Year :	SFH :	No. of Courses :	Product :
Last Inspection Date :		Current Inspection Date :	

INTERNAL INSPECTION CHECKS

1.0	Internal Condition	Observations	Recommendation
1.1	Bottom plate painting		
1.2	Shell plates painting		
1.3	Visual Condition of weld seams of bottom and shell		
1.4	Shell and bottom plates corrosion and pitting.		
1.5	Shell to bottom plate joint		
1.6	Edge settlement		
1.7	Bottom plate thickness readings		
1.8	Sump plate thickness readings		
1.9	Water draw-off line		
1.10	Roof Support Columns		
1.11	Reinforcement pad conditions in bottom plates for roof support columns		
1.12	Structural under roof plates viz. rafters, fasteners, gussets etc.		
1.13	Steam coils thickness, supporting and hydraulic test.		
1.14	Nozzles Thickness		

2.0 Any other observations:

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ANNEXURE - II**SAFETY IN INSPECTION OF STORAGE TANKS**

1. A person shall not go alone on the roof of storage tanks for inspection. For repair/ maintenance/ testing, the entry to the roof deck shall be allowed through a work permit.

No walking shall be done on the roof plates where thinning/ perforations are observed. Necessary planking may be provided for walking on such roofs.

Ultrasonic thickness measurements shall not be carried out when the tanks is receiving or dispatching.

Entry to deck of a floating roof shall be allowed with related precautions when the floating roof is more than 5 meters below the top angle ring.

2. H₂S may be present inside the storage tanks in hydrocarbon service. In case of floating roof tanks, H₂S may be present above roof also. Necessary safety precautions shall be taken while inspecting such tanks.

3. No hot work shall be carried out inside the tank prior to gas freeing of the same.

Hot work on bottom plates shall be done under strict supervision where perforation and/ or leaks are observed.

Air test pressure shall NEVER exceed the specified limits.

4. All leaded tanks must be completely shot blasted and thoroughly cleaned before entering. API publication No. 2015, & 2202 shall be referred for the storage tanks containing leaded petroleum products.
5. Safety belts shall be worn while working on Bosun's chair.
6. Proper ventilation arrangement should be made for shot blasting, hot jobs, painting etc

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ANNEXURE - III**HYDROSTATIC TEST OF STORAGE TANKS**

This annexure outlines recommended procedure for conducting hydrostatic testing of storage tank.

Pipe connections: No pipes should be connected to the newly built tank during the water test except the one used to fill the tank with water. Product lines should not be connected to the tank until the completion of hydrostatic test to ensure that they are not affected by any settlement occurring during the test. Care should be taken on piping connections while hydro testing especially after a major pad/ foundation repair.

Water Quality: Fresh clean water should be used for the hydrostatic test if possible. At locations where clean water is not available in sufficient quality or where the cost of using such water is prohibitive, river water, dock water or seawater may be used. In such cases, a water chemistry test should be performed to determine pH, chloride content and the presence of other potentially corrosive substances. The analysis will provide an indication of the corrosiveness of the water. A pH of between 6 and 8.3 is acceptable where the lower or higher pH would increase corrosiveness.

Whenever unclean water is used for testing, the length of time that the water is kept in the tank should be kept as short as possible. The corrosiveness of the water will determine the maximum time that should remain in the tank. If the time limit is unacceptable, corrosion inhibitors may need to be added to the water. Immediately after draining the test water, the tank interior should be hosed down using fresh, clean water.

Where aluminium internal floating cover is fitted in the tank, it is necessary that clean water be used.

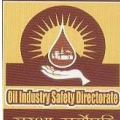
Test Temperature: The hydro test subjects the tank to its heaviest load. Locations with stress peaks can yield. The ability to accommodate yielding is influenced by the toughness of the material. The toughness is lowest at low temperatures. It is therefore essential that hydrostatic testing be carried out at a temperature of 4 Degree centigrade and above. Suitable considerations should be given to the applicable material specification and design parameters while deciding minimum test temperature

FILLING RATES:**General:**

Filling rates and the need for intermediate hold periods should take into account the soil condition and foundation design. The tanks built on a stable foundation with no predicted settlement eg. rock or a piled concrete slab can be filled at a rate of up to 1.5M/hour, while taking into account other parameters like roof venting capacity.. Same rate of filling can be used during repeat hydrostatic test after a major repair of tank where the foundation has already undergone the settlement cycle and no additional settlement is expected.

Filling rates should be limited to allow the subsoil layer under the tank to absorb the settlement without problem in the following cases;

- Tanks built on foundations where significant settlement during filling is predicted
- In service tanks where major pad repairs have been carried out due to which significant settlement during filling is predicted.

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The filling rates, the holding time and the frequency of settlement monitoring should be laid down in the hydrostatic test procedure.

When filling a fixed roof tank, it is necessary to ensure that sufficient venting capacity is available. It is usually adequate if all the roof vents plus at least one roof manhole are fully open. When filling a tank having external floating roof or internal floating roof, a low filling rate of not more than 0.25m/hour should be used until the roof is floated. Thereafter, the filling rate may be increased to the specified maximum filling rate.

Filling Procedure:

It is preferable that a written filling procedure is developed for each tank, based on the soil conditions and foundation design. The procedure should outline the filling rates and holding time between each stage.

A generalized filling procedure is given below, which needs to be re-confirmed before commencing the hydrostatic test. Monitoring during filling and holding periods for each stage is covered separately under settlement monitoring.

Stage 1:

- Fill the tank to 0.5 H (height of the tank) at 70% of the design product-filling rate, as per tank design conditions.
- At the end of stage 1, there should be a holding period of settlement / monitoring (recommended 24 hours minimum) before the commencement of stage 2.

Stage 2:

- Recommended filling to 0.67 H at the same filling rate as above.
- At the end of stage 2, there should be a holding period of settlement / monitoring (recommended 24 hours minimum) before the commencement of stage 3.

Stage 3:

- Recommended filling to 0.83 H at the same filling rate as above.
- At the end of stage 3, there should be a holding period of settlement / monitoring (recommended 24 hours minimum) before the commencement of stage 4.

Stage 4:

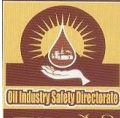
- Fill to H at the same filling rate as above

At the end of stage 4, the tank should be left full and monitored in accordance with the holding time detailed below.

The sequence of filling various tanks in a tank farm shall be such as to avoid over-lapping of settlements. To achieve this, it would be necessary to fill diagonally opposite tanks when simultaneous or individual filling is carried out

Maximum Filling Height:

The tank should be filled to the overflow nozzle designed to limit the liquid level or to the maximum design filling height. The tank should not be over filled. Overfilling of a fixed roof tank

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will cause an upward pressure against the underside of the roof plate, which could result in the top angle and roof supporting trusses being buckled. Overfilling a tank with an internal floating roof can seriously damage and possibly sink the internal roof.

Holding Time:

For tanks with no settlement, a minimum holding period of 24 hours is recommended. This is the time required for small leaks to show up. If there is rain during the holding period, the period may need to be extended to observe the tank under dry conditions.

For tanks with predicted settlement, the holding time should be 48 hours or longer until settlement monitoring gives a clear indication that the rate of settlement is diminishing.

Emptying after Hydro Test:

When the hydro test is finished, emptying under gravity is sometimes used to reduce the time taken. The vacuum relief capacity of the roof vents of fixed roof tanks should be checked to ensure it is adequate for the proposed emptying rate. Vacuum relief capacity can be increased by opening top manholes

When withdrawing water from an external floating roof tank, it should be ensured that the bleeder vents are operating properly. Also, the withdrawal rate needs to be reduced when the roof approaches its landing position. Floating roofs should be landed gently to avoid damage. The same applies to fixed roof tanks with internal floating roofs.

Settlement Monitoring:

A settlement-monitoring programme is an essential part of the hydraulic test. Monitoring frequency depends on soil conditions and expected settlement.

Shell level measurements should be taken with the tank empty, 50% full, 67% full, 100% full and when empty again. For tanks with predicted settlement, the monitoring frequency shown in attached table may be employed. The stages referred in the table are as defined in above paragraph.

Shell settlement readings should be taken at all reference points along the circumference. The readings should be evaluated by calculating the total tilt and the deviation from the uniform tilted plane. API-653 "Tank Inspection, Repair, Alteration and Re-construction", gives guidance on settlement evaluation.

Measurement of tank bottom profile should be done before starting to fill the tank and again when the tank is nearly empty. When draining the test water, leave a level of about 0.25m in the tank to ensure that the tank bottom is in full contact with its foundation profile. The tank bottom profile should then be measured and evaluated.

A	Before filling starts	Tank levels around shell circumference
B	With 0.25m water in tank	Measure bottom profile, where significant settlement is predicted
C	During filling to stage 1	Take levels every 24 hours
D	While holding at stage 1	Take levels every 12 hours until rate of settlement diminishes

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E	During filling to stage 2	Take levels every 12 hours
F	While holding at stage 2	Take levels every 12 hours until rate of settlement diminishes
G	During filling to stage 3	Take levels every 12 hours or more frequently depending on rate of settlement
H	While holding at stage 3	Take levels every 12 hours until rate of settlement diminishes
I	During filling to stage 4 (full)	Take levels every 12 hours or more frequently depending on rate of settlement
J	While holding at stage 4 (full)	Take levels every 12 hours until rate of settlement diminishes
K	Nearly empty with 0.25m water remaining	Measure bottom profile to establish total bottom settlement, where significant settlement has been observed
L	Completely empty	Take levels to establish total shell settlement

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Annexure-IV

A Typical List Of Tools Required For Tank Inspection

- i) Ultrasonic Thickness Meter/ Flaw Detector
- ii) Radiographic Equipment
- iii) Dye Penetrate Kit
- iv) Holiday Detector
- v) Shore Hardness Meter
- vi) Paint Thickness Gauge
- vii) Vacuum Box Tester
- viii) Safety Torch
- ix) Knife/Scraper
- x) Crayon
- xi) Magnifying Glass
- xii) Permanent Magnet
- xiii) Vernier Caliper/ Micrometer
- xiv) Fillet Gauge
- xv) Measuring Tape/ Scale
- xvi) Straight Edge
- xvii) Pit Gauge
- xviii) Plump Bob and Line
- xix) Magnetic Particle Testing Kit
- xx) Floor Scanner (MFL, LFET, UT, ECT)
- xxi) Manometer
- xxii) Oil Chalk Test Kit
- xxiii) Hydraulic Test Pump
- xxiv) Tank Crawler for UT Gauging
- xxv) Acoustic Emission Test Kit
- xxvi) Digital Camera
- xxvii) Half Cell
- xxviii) Binocular
- xxix) Spring Balance
- xxx) Hardness Tester

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Annexure-V**CHECKLIST FOR PRE COMMISSIONING INSPECTION**

TANK NO.

DATE OF INSPECTION

DRG. NO.

LOCATION

PRODUCT STORED

TYPE OF TANK

CAPACITY

MAIN DIMENSION

DIA

HT.

ERECTION CONTRACTOR

NAME OF INSPECTION AGENCY/INSPECTOR

CHECK LIST

1. Foundation pad/ anchor bolts.
2. Settlement of foundations.
3. PWHT for clean out doors and nozzles wherever applicable.
4. Shell for obvious abnormalities in respect of out-of-roundness, bulges, dents, etc.
5. Alterations made during construction have been incorporated in as-built drawing.
6. Measure and record wall thickness of shell, bottom, roof & Nozzles
7. Provision of Non-return valve at roof drain opening has been provided (for floating roof tank)
8. Proper installation of floating roof seals
9. Proper fabrication of roof bleeder vent
10. For floating roof tanks, check foam dam, foam system, sprinkler system. For fixed roof tanks, check the functioning of sprinkler system, foam pourer lines and its connections with the tank.
11. Internals, steam coils, float gauge, roof drains, etc.
12. Nozzle facings, gaskets and bolts
13. Insulation

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14. Painting quality and coat thickness
15. Relief Valves, Pressure-Vacuum (PV) Valves, mixers and other tank mountings
16. Emergency roof drains water seal
17. Internal cleanliness before final boxing-up.
- 18 Tank earthing and earthing pit/ grid resistance. For floating roof tanks check for electrical continuity between shell and floating roof.
19. Cathodic protection system
20. Availability of the stenciling with information like tank number, capacity, service, last inspection date etc..

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Annexure-VI

LIKELY TYPES & AREAS OF METAL CORROSION/ WASTAGE IN TANKS

Metal wastage in a petroleum storage tanks generally occurs due to one or more of the following:

- i) Sea water corrosion
- ii) Chemical corrosion (i.e. Sulphur)
- iii) Vapour corrosion
- iv) Atmospheric corrosion
- v) Bacterial corrosion
- vi) Stress corrosion
- vii) Soil corrosion
- viii) Combination of Erosion & corrosion

A storage tank shall be protected from Corrosion so that metal loss is kept to a bare minimum.

A. BOTTOM PLATES

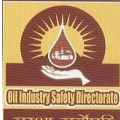
Corrosion in bottom plates is found from both fluid side and soil side. It gets corroded rapidly if the fluid stored is having water content. If the water happens to be seawater then corrosion takes place at a faster rate.

This is further aggravated, if there is undulation in the tank bottom plates, where grooving type general corrosion is observed.

Bacterial corrosion of the bottom plates is generally observed in Crude and HSD tanks having high sulphur content. The bottom plates develop deep isolated pits, which eventually puncture and the bottom starts leaking. Weld failures are observed in the bottom plates where the tank is in caustic service.

The projecting out portion of bottom plates is prone to corrosion at the edges due to seepage/ accumulation of water between the foundation and the bottom plates as a result of soil accumulation caused by settlement of tank pad. Due to creation of voids in between the foundation and the tank bottom, there is possibility of accelerated underside corrosion. The bottom plate is prone to wear due to gauging.

The bottom plates i.e. annular and sketch plates in front of the tank inlet nozzle/ tank mixers get damaged specially in high Sulphur crude storage tanks and where sludge accumulation is high due to a combination of EROSION and CORROSION. Initially corrosive compounds present in crude oil attack bottom plates and forms a loose corrosion product/ scales. These loose product get easily removed/ washed out by impingement of crude oil during receipt operations, normally at very high flow rate. This cyclic process of Corrosion & Erosion causes the tank bottom plate thinning.

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The underneath surface of bottom plates get corroded due to corrosive soil mainly due to ingress of corrosive water/sediment during tank bottom leak/seepage.

The bottom plates below the roof leg resting location are also vulnerable for corrosion/ damages.

B. SHELL PLATES

Shell plates generally get corroded internally where liquid/ vapour phase is maintained (i.e. middle to top shell courses). Internal corrosion in the vapour space is most commonly caused by hydrogen sulphide vapour, water vapour and oxygen, giving pitting type corrosion. The bottom shell course gets corroded in bottom 300 mm height when the tank contains water in its product.

In floating roof tank, inside surface of shell plates approx. 1-2 meters from top of shell gets corroded due to combination of atmospheric corrosion & presence of product vapour

Corrosion can occur on all external parts of the tank, particularly if there is water entrapment in horizontal parts like wind girder, annular platform supports etc. and is high in coastal atmosphere. This type of corrosion may range from negligible to severe depending upon the atmospheric conditions of the locality.

In case of insulated tanks, corrosion of shell plates is observed where insulation gets damaged & water seepage takes place. The corrosion in such tanks are more predominant in the bottom most shell course and the extended portion of annular plate

Stress corrosion cracking (Caustic Embitterment) can be a problem in services like Caustic/ MEA/ DEA. The attack is severe on the bottom shell courses.

C. FIXED ROOF PLATES /STRUCTURAL

Underside of the roof and roof structural come under corrosive attack due to vapour corrosion in tanks storing FO, Asphalt & HSD. The contact areas between roof plates and structures are prone to heavy attack due to crevice corrosion and thinning is observed in these areas.

In case of insulated tanks, corrosion of roof plates is observed where insulation gets damaged & water seepage takes place. The corrosion in such tanks are more predominant where undulation in roof plate is more.

Atmospheric corrosion can occur on all external parts of the tank and is high in coastal atmosphere. This type of corrosion may range from negligible to severe depending upon the atmospheric conditions of the locality.

D. FLOATING ROOF

i) Floating Roof Plates (Deck Plates)

Floating roof deck plates are prone to corrosion due to rainwater accumulation on the deck. The corrosion is predominant at locations where undulation, in roof plates, is more. Underside of the roof gets corroded where vapour pockets are formed. Atmospheric corrosion can occur on all external parts of the tank and is high in coastal atmosphere

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ii) Pontoon Boxes

Pontoon boxes on the floating roof are prone to corrosion at the fillet weld between the pontoon and deck plates.

iii) Rim Plates

Rim plates at the outer periphery of the floating roof get corroded where liquid-vapour phase is maintained. This is approximately at the center of the rim plate.

iv) Roof Legs/ Assembly

Roof legs get severely corroded at the liquid-vapour phase junction. The roof leg sleeve gets corroded near fillet weld junction to the roof plates and at the bolt hole area. Roof sleeve pad may get corroded at the underside if sealing run between roof plates and pad is not carried out.

v) Roof Drain Sump

Roof drain sump gets corroded due to accumulation and stagnation of water in the event of blockage with debris.

E) WATER DRAW NOZZLES/ PIPES

The water draw-off nozzle pipes get corroded due to water stagnation.

F) STEAM COILS

Steam coils get thinned out at the bends due to internal erosion and stagnant condensate. Leg supports of steam coil get corroded at the bottom portion due to accumulation of water. Steam coil supports get dislodged due to thermal expansion. External corrosion of steam coil is more where water accumulation & sludge is more in the tank

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Annexure-VII**CALCULATION OF REJECTION LIMITS FOR SHELL PLATES****1. General**

In determining the limiting thickness for the shell plates of a tank, either for the purpose of pre-calculating a set of retiring thickness for each tank or as a matter of necessity at the time of inspection, the basic method given in the applicable standard shall be used. The result will be a thickness which will be the minimum required for a particular location for the given tank. When that thickness is reached, repairs or replacement shall be required.

A pit or a very small area reduced to the retiring thickness is however taken as not weakening the plate appreciably from the standpoint of resisting pressure. For determining the average/ retiring thickness in each shell course when there are corroded areas of considerable size and repair methodology, guidelines given in API –STD- 653 shall be followed

2. Arbitrary Limits For Top Shell Courses

The rejection limit for shell plates, as specified under clause (1) shall also be applied for the top shell courses, but these courses shall in any case be rejected when due to corrosion the plate thickness has reached, over a considerable area the following value:

- a. When the original thickness was 6 mm: 2.5 mm.
- b. When the original thickness was 8 mm: 3.2 mm.

It is pointed out that for many tanks, especially of medium and large size, the top shell courses may buckle before the limits mentioned under (a) & (b) have been reached. The rejection limit for top shell courses shall therefore, not be determined before the stability of the shell has been checked according to the requirements described under clause (3). All repairs shall be as per API - 653.

3. Buckling Of Upper Shell Courses When Tank Is Empty**a) General**

The rejection limits for the shell plates specified in clause (1) and clause (2) are based on the condition that the tank is completely filled with liquid. However, when the shell plates have corroded it may be possible that buckling of plates occurs before the above-mentioned limits.

Buckling of shell plates will always occur in the upper half of the tank shell, as the upper courses are thinner than the lower courses.

Shell distortion include out of roundness, buckling, flat spots and peaking & banding at weld joints. Shell distortion can be caused by many conditions such as foundation settlement, over or under pressure, high wind, deficiencies in fabrication & repairs etc.

b) Loading Conditions

Buckling of shell plates may occur when the stability of the tank shell is insufficient to withstand one or the combination of the following loads:

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- (a) Wind on the outside of the tank shell and for open top tanks the wind load on the inside of the tank shall also be considered.
- (b) Vacuum inside the tank
- (c) Dead load of roof and supporting structures.
- d) Seismic loads.
- e) Operation at temperature over 200 deg. F
- f) External loads caused by piping, tank mounted equipment, hold down lugs etc.
- g) Loads due to settlement

c) Method of Calculations

The stability of the corroded tank shell against buckling shall be controlled in accordance with the calculation method given in BS-2654-part 3 'Higher Design Stresses'.

It is then necessary to specify the average thickness of the corroded shell courses, especially of the upper half of the tank shell.

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Annexure-VIII**TANK REPAIR AND INSPECTION****1. ROOF REPAIR****1.1 Roof Replacement**

Entire or partial replacement of corroded roof plates shall be done with new plates of thickness as provided in the original design. After replacement the welding of the roof plates shall be checked for leaks using vacuum box. The fixed roof may be tested by applying internal air pressure also. The internal air pressure shall not exceed the weight of the roof plates or 75 mm of water column. After application of air pressure, the joints are checked by soap solution. For the deck plate weld joints of the floating roof, oil penetration test may be used as an alternative to vacuum box test.

The roof support structural shall be repaired such that the design conditions are restored. The roof integrity may also be checked by creating internal vacuum inside fixed roof tanks. It is current practice to test the roof after construction at vacuum of 25 mm of water column. Vacuum is produced by controlled draining of water after hydrostatic test. The vacuum test shall be done with utmost care and in no case shall vacuum exceed 25 mm of water column.

In the floating roof tanks, new pontoon box welding should be checked by air and soap solution. Alternatively, these may be checked by floating the roof of the tank with water and checking the pontoon compartments for any leak. The tank roof shall be floated on water after completing major repairs.

1.2 Weld Repair

The weld repairs should be carried out by gouging/ grinding the leaky spot and welding. Repairs shall be inspected for their integrity by vacuum box test/ air test.

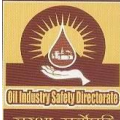
1.3 Repair to Roof when Tank is in Service

When deep pits in tank plates are not closely spaced and extensive, and thus do not affect the strength of the tank, they can be repaired by other methods if welding is not practicable. Any methods that will stop the corrosion and plug the leaks will be satisfactory. Filling with proprietary air-hardening adhesive may be suitable if it will not be affected by the tank contents. Any other material of putty like nature that hardens upon drying should be used only for temporary repair. Such material must be able to tolerate the tank contents in addition to making a tight bond with the steel plate. In all cases the pits shall be cleaned thoroughly.

Leaks in the roof are commonly repaired by providing "Soft Patches" that do not involve cutting, welding, riveting or bolting. The soft patches can be made from a variety of materials including canvas, asbestos, rubber, neoprene, glass cloth, FRP, FRE, Asphalt and Proprietary mastic or Plastic sealing material, the choice depending upon the contents of the tank and service conditions. The patches may be applied when tank is in service.

The above are temporary methods of repair. Proper and permanent repairs shall be carried out at the earliest opportunity.

In the case of foam seals, in-service seal repairs are normally limited to repair or replacement of secondary seals/ weather protector.

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In the case of rim mounted shoe seals, the primary seal fabric can be replaced in-service taking precautions.

In case of liquid filled seals, the leak in the seal tube can be repaired in service. When the seal is replaced or reinstalled the roof shall be checked for floatation

1.2 SHELL PLATE REPAIR

1.2.1 Shell Replacement

Complete or partial replacement of corroded and thinned out shell plates shall be done using new plates of thickness as provided in the original design. Partial replacement of shell plate can be done by cutting window at the affected portion. The corners shall be rounded off. After welding the joints shall be checked visually or by Penetrant Test (PT) and taking spot radiograph. After satisfactory repairs, the joint shall be checked for leaks by filling the tanks with water as is done in the case of newly fabricated tanks.

1.2.2 Weld Repair

Defective welds (originally present or developed in service) or leaky welds shall be repaired by gouging, grinding and welding. The welding shall be thoroughly inspected by magnifying glass or by PT. Spot radiographs should also be taken. The hydrostatic test of the tank shall be done after major weld repairs in line with relevant standard

1.3 NOZZLES REPAIR

The thinned nozzles shall be taken out by gouging the welding. The new welding shall be checked with air at a pressure of 1.06 kg/cm^2 and soapsuds through the telltale hole in reinforcing pads. In case tank is to be hydrostatically tested, the nozzle-reinforcing pad shall be checked for any leakage through the telltale hole. If leakage is observed, the inner welding of nozzle with shell shall be gouged, re-welded and tested. Addition or modification of nozzles shall follow the requirements of the applicable construction code

1.4 TANK BOTTOM REPAIR

1.4.1 Bottom Replacement

Partial or complete replacement of tank bottom plates can be done using plates of thickness provided in the original design. The replacement plates can be taken in to the tank through window cut in bottom shell course. A new bottom may be laid on the old bottom when it is not possible to take out the old plates. This arrangement has been shown in Fig. No1.4.1.1, 1.4.1.2 and 1.4.1.3.. Laying of new bottom plates directly over the old bottom plates may be avoided. Nozzles and heating coils should be re-positioned on the renewed bottom as per the operational requirement.

The new weld joints shall be checked thoroughly. To detect any leakage in the weld joints, vacuum box test shall be carried out.

Alternatively, the bottom plates can be checked pneumatically. Air connections are installed at minimum of 5 points. An air pressure of 10 cm. water column is maintained with a centrifugal blower or an air compressor. A U-tube manometer indicates this pressure and at the same time safeguards the bottom against an over-pressure. The entire bottom is then tested.

This system has an advantage over the vacuum box method as it is much faster and more secure since the entire surface and not only the welded seams are covered. Another advantage is that

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while the air pressure is applied, the bottom is partially lifted from its base. This causes the metal plates to flex so that possible holes of minor dimensions which might have been sealed by soil or rust particles will become detectable.

If an air pressure of 10 cm. water column cannot be maintained owing to too much air leakage around the outer rim of the bottom, it is recommended to install a dam around the tank as close as practicable to the shell wall. A dam of clay or bitumen sand mixture about 20 cm. high should be applied. This dam has to be sealed with a bitumen layer. The space between the dam and the tank wall is then filled with fresh water, which acts as seal. Arrangement of tank bottom testing is given Fig 1.4.1.3.

However it should be noted that pneumatic testing of the bottom plates as described above will not give correct results if the air is not able to reach entire area beneath the bottom plates and entrapped in pockets.

Shell to bottom weld joint shall be tested by filling the tank with water to a level of half the tank height.

The critical zone for repairs on the tank bottom is within 3 inch of the inside edge of the shell, measured radially inside. No welding or weld patch are permitted in this critical zone except for welding of deep scattered pits, crack in the bottom plate; the shell to bottom weld or where the bottom or annular plate is being replaced.

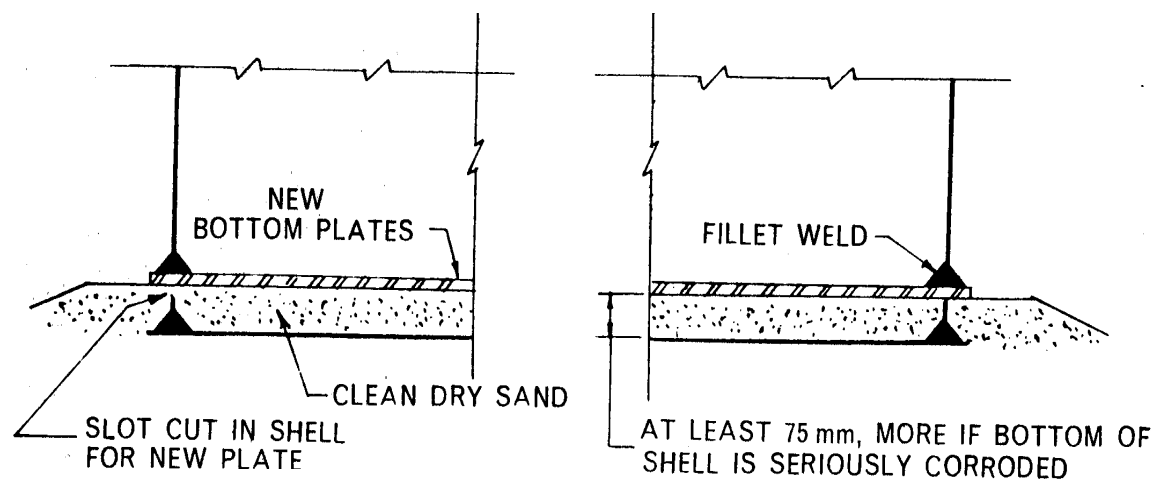


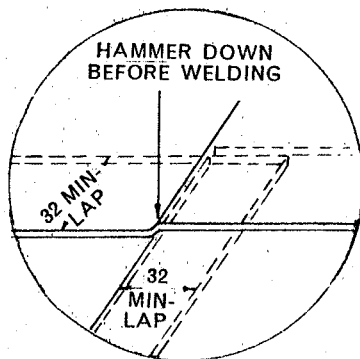
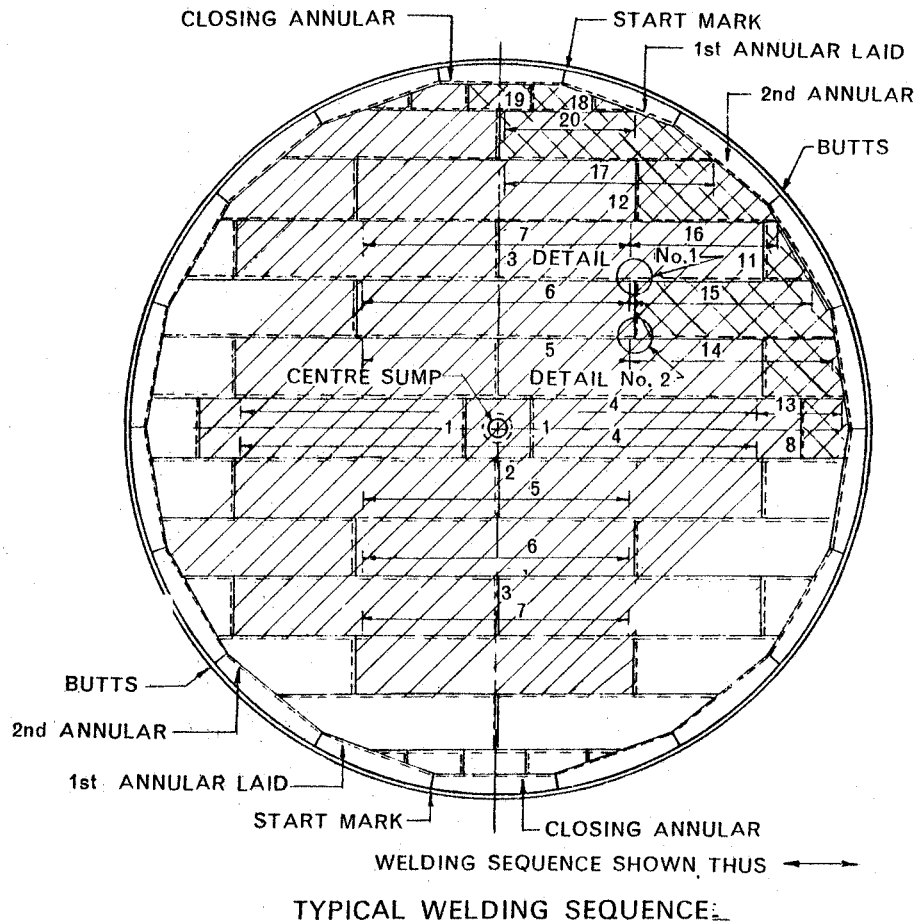
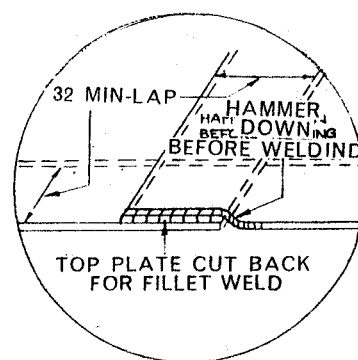
Fig 1.4.1.1

Fig. 4-A :METHOD OF REPAIRING TANK BOTTOMS



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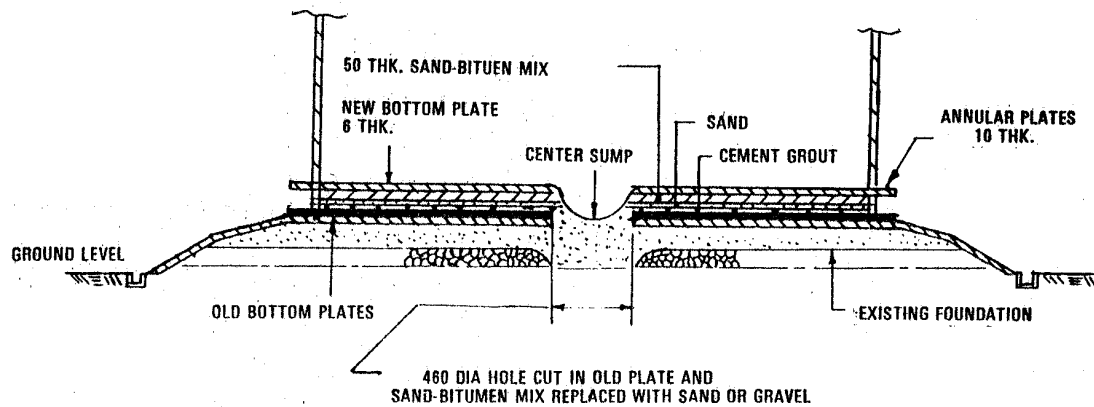
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**DETAIL No. 1**JOINT AT JUNCTION OF 3 PLATES
LAP OVER TOP PLATE**DETAIL No. 2**JOINT AT JUNCTION OF 3 PLATES
LAP OVER BOTTOM PLATE**Fig 1.4.1.2 :FITTING NEW TANK BOTTOMS WELDING SEQUENCE**

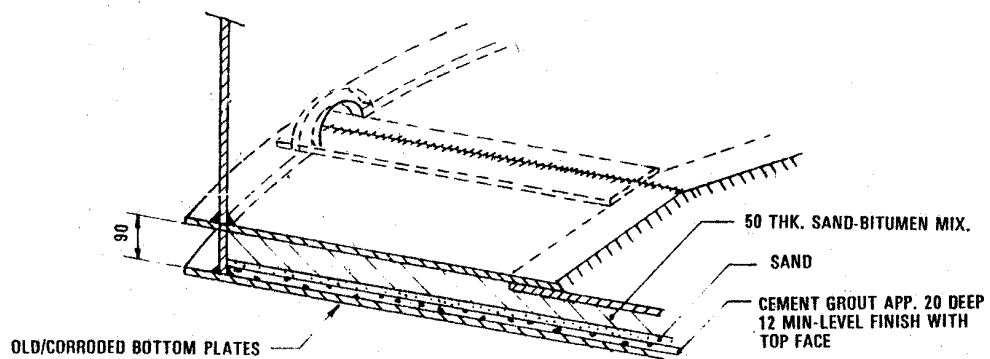


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SECTION SHOWING FILLING UNDER NEW BOTTOM



PART SECTION SHOWING NEW BOTTOM PLATES

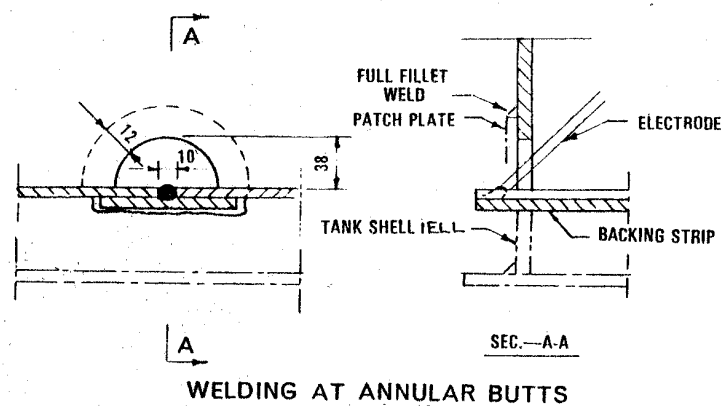
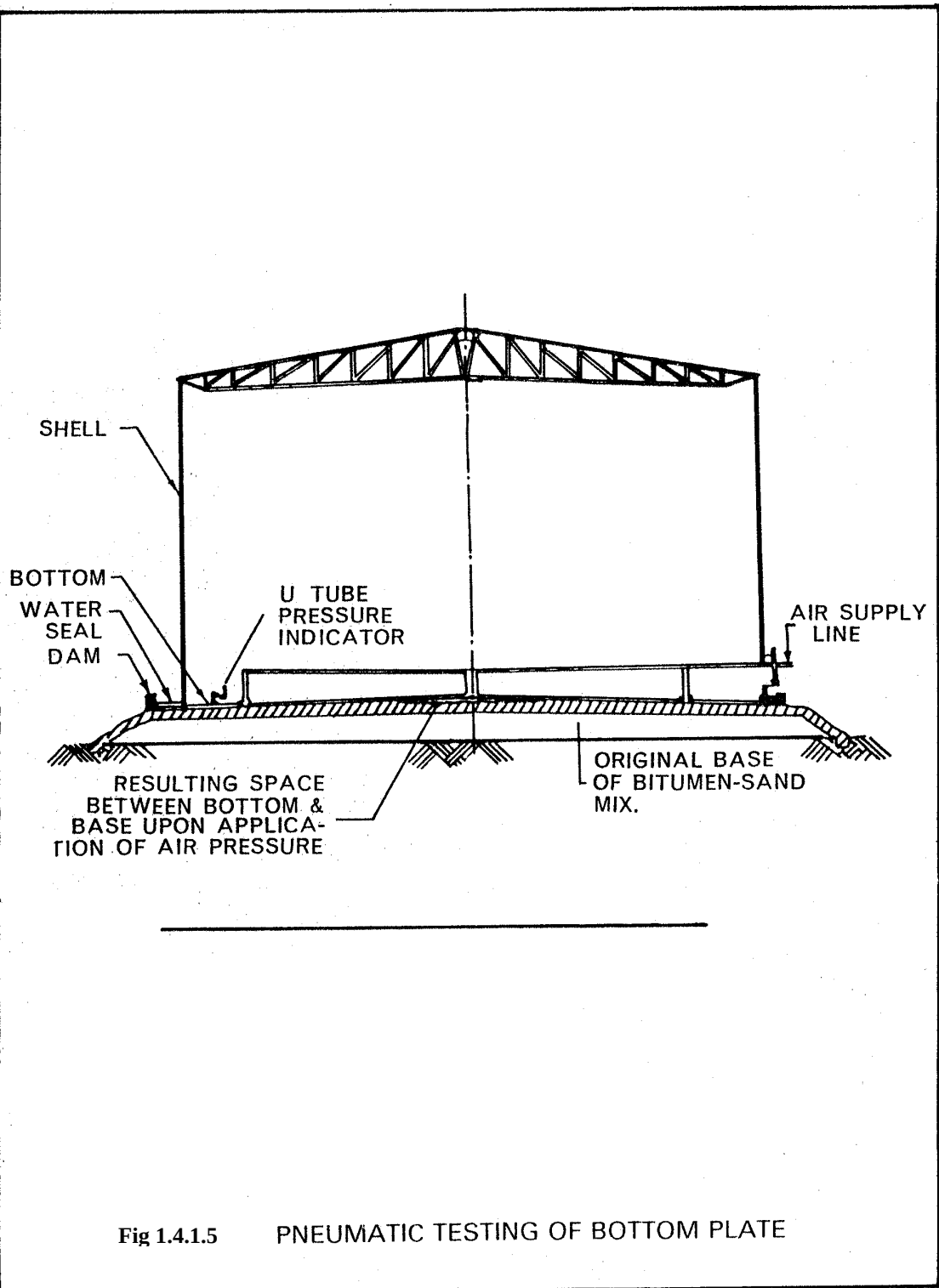


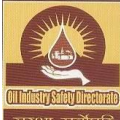
Fig 1.4.1.3 : FITTING NEW TANK BOTTOMS



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1.4.2 Weld Repair

Leaky welds in the bottom can be repaired by gouging, grinding and welding. Repaired welds shall be checked by vacuum box testing for lap joints and spot radiograph for butt joints.

1.5 STEAM COIL REPAIR

Thinned or deteriorated steam coils shall be replaced. The weld joints shall be checked by spot radiography. After satisfactory repairs, the coils shall be hydrostatically tested at a pressure of 1.5 times the operating pressure.

1.6 TANKS PAD REPAIR

1.6.1 Erosion

The raised foundation of vertical tanks must be protected from the effects of erosion. Any damage to the surface of the sealing coat or any breakdown of the sand-bitumen mix of that part of the foundation which project beyond the base of the tank shall be repaired before the underlying foundation is damaged.

1.6.2 Settlement

Even with relatively minor settlement, the outer edge of the bottom plates of a vertical tank will settle at a level below the surface of the sealing layer of the foundation. This results in the formation of a channel around the periphery of the tank, in which rainwater collects. When this occurs, small outlet channels in radial direction shall be cut in the sealing layer of sand bitumen mix/ cement at the lowest point and at intervals of about 6mt. around the periphery to provide drainage. The relative settlement shall be checked. If settlement exceeds 25 mm, this method may destroy the effectiveness of the sand bitumen-sealing layer. In such cases, the surface of the projecting part of the foundation shall be trimmed and a new sealing layer of sand bitumen mix. 50 mm thick should be laid to provide a proper drainage with a surface sloping away from the toe of the tank bottom.

1.6.3 Tilting

The maximum allowable tilting in the fixed roof tanks due to uneven settlement shall be as shown in Fig 6.

The maximum allowable tilting in floating roof tanks will be governed by the designed range of gap between roof and shell.

The maximum allowable tilting as given Fig 1.6.3.1 will cause an increase in the hoop stress of the shell plates by 2% of the hoop stress calculated for tanks without uneven settlement. This increase shall be deducted from the allowable stress when calculating rejection limit of shell plates of tanks, which have settled unevenly.

A tank shall be lifted and the foundation repacked if the limit for tilting is reached.



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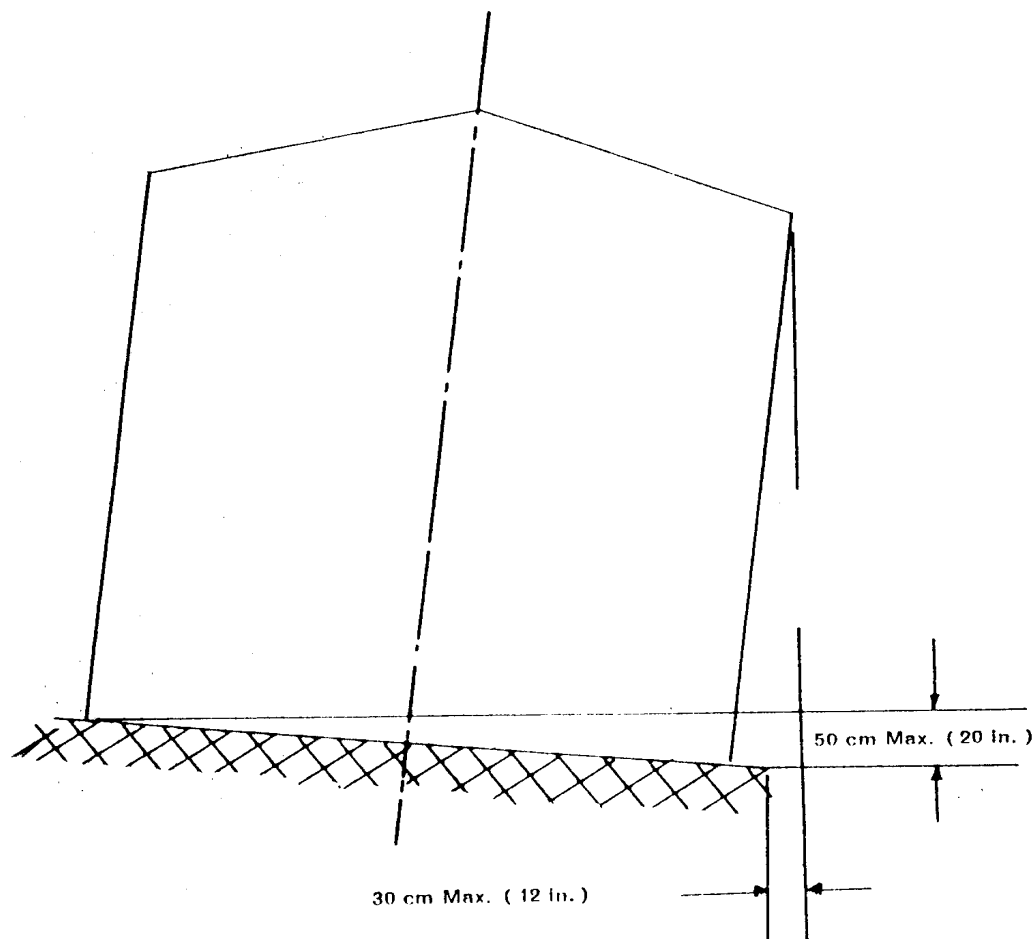


Fig 1.6.3.1 MAXIMUM ALLOWABLE TILTING DUE TO UNEVEN SETTLEMENT